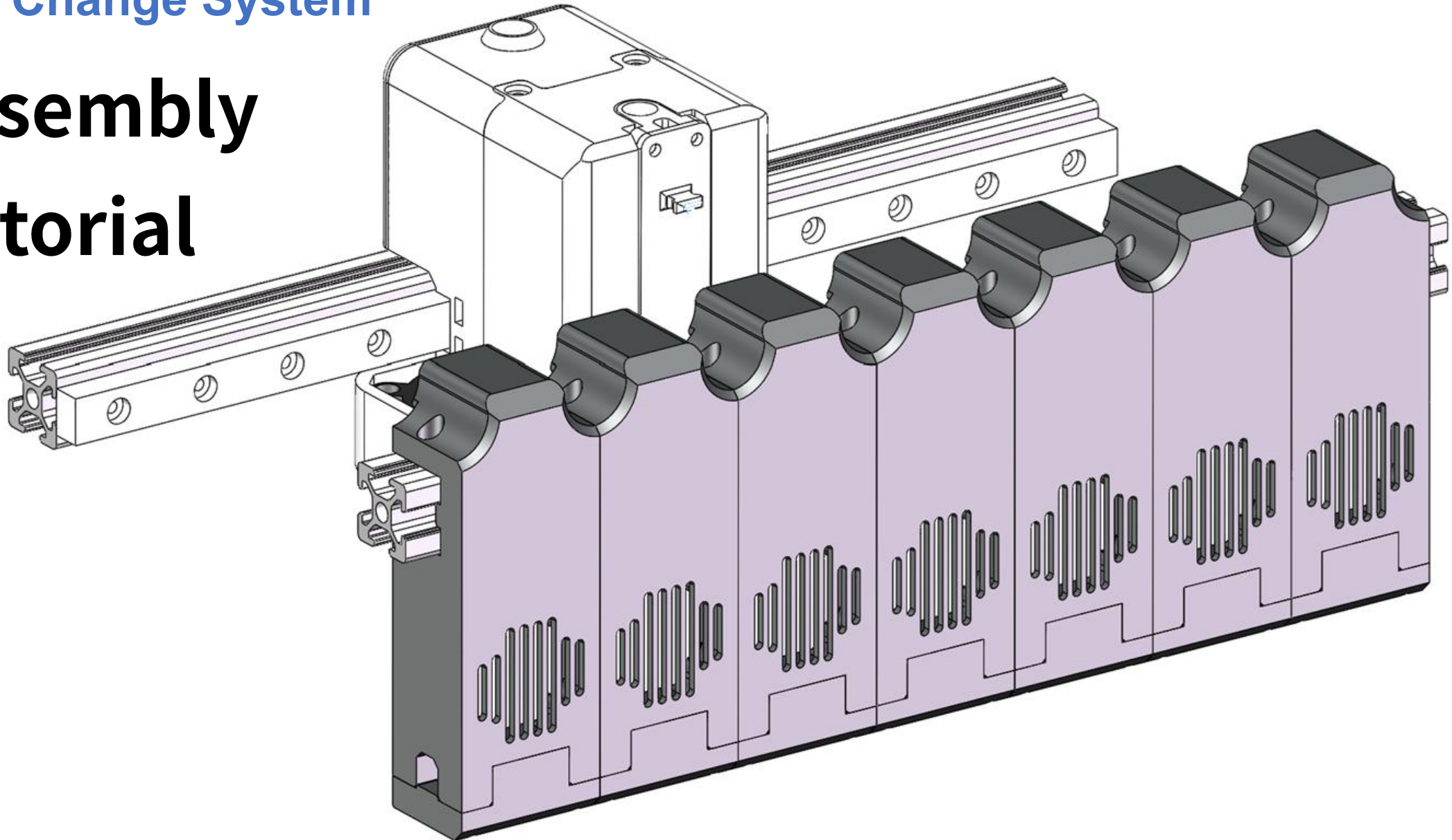
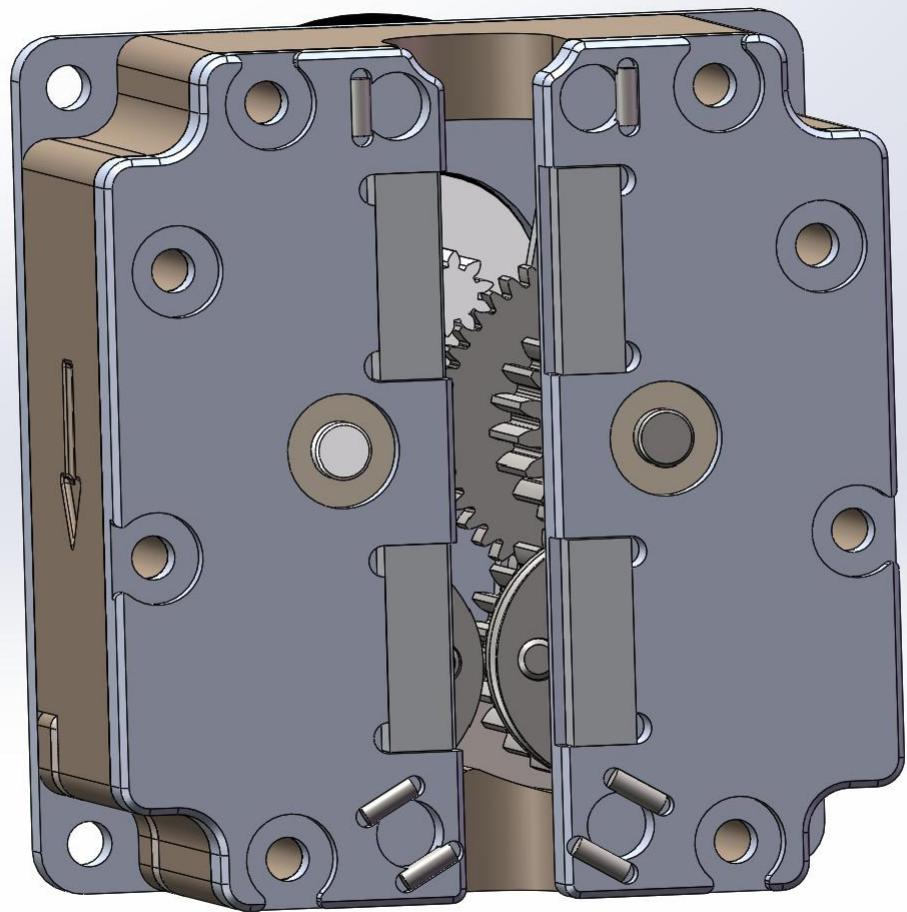


“YUDX”
Tool Change System

Assembly Tutorial



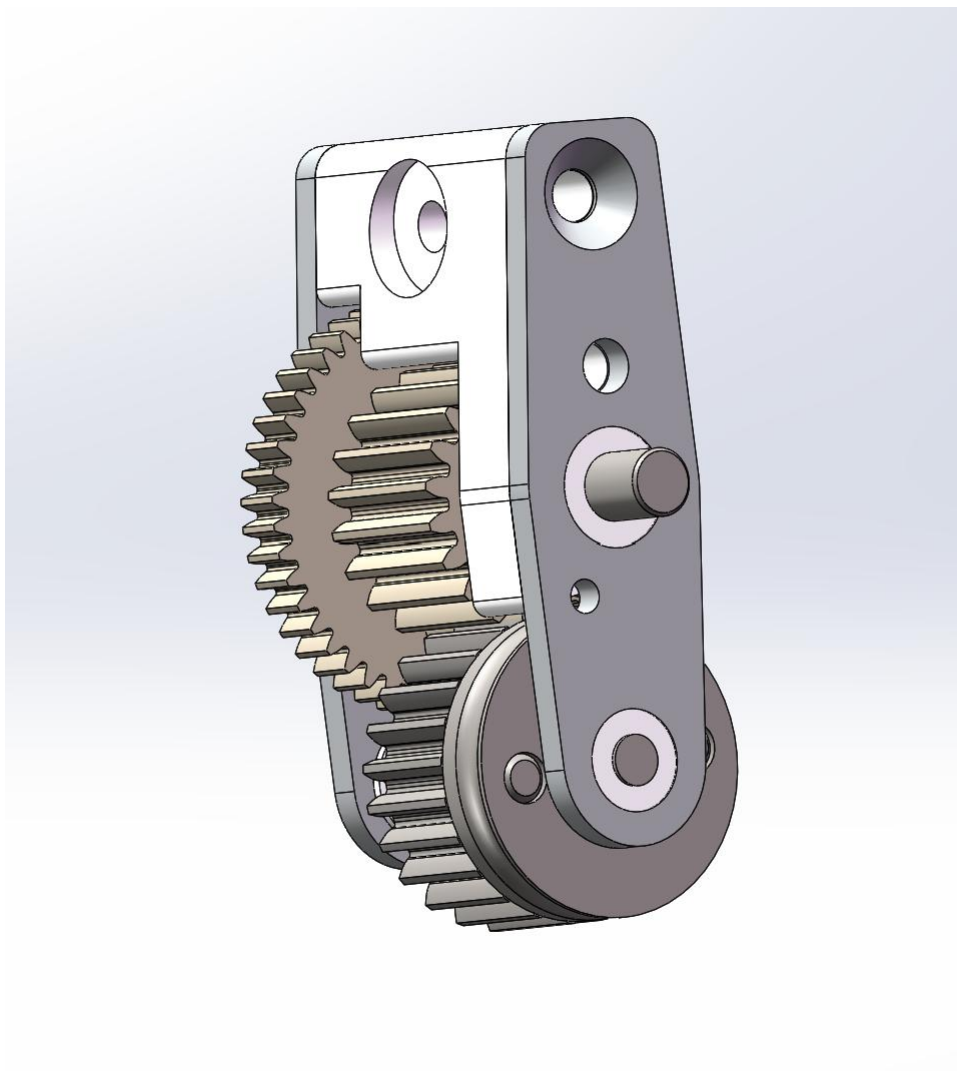
Part 1: Extruder Assembly



Extruder

If you purchased the extruder finished kit, you can skip this step. This is what the extruder looks like after assembly.

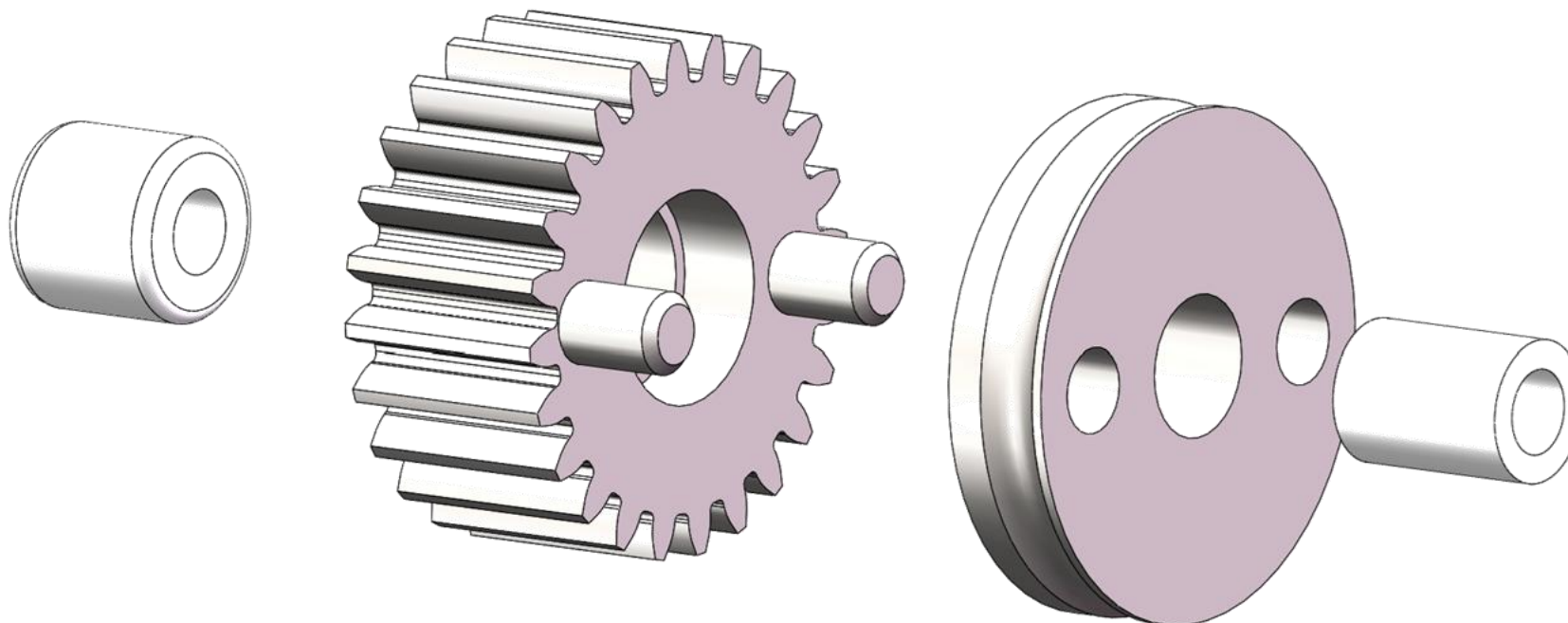
01 Assemble the Metamorphic Gear Set



Metamorphic gear set

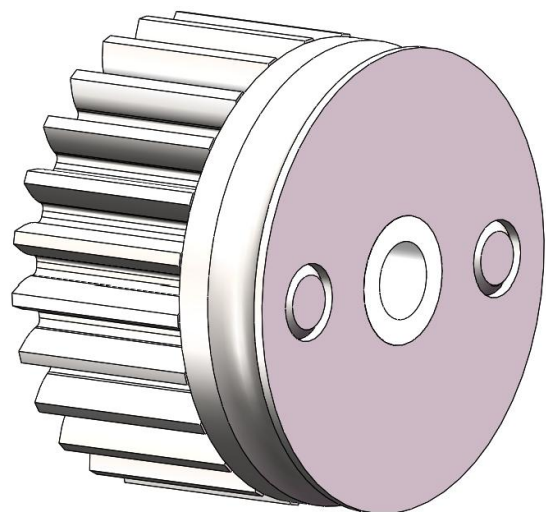
This is what the gear set looks like after assembly. The diagram shows the right-side gear set. Since the left and right sides require two completely mirrored metamorphic gear sets, it is recommended to assemble both at the same time. The tutorial uses the right-side gear set as an example. Many parts of the gear set mechanism are easy to install incorrectly or reversed, so be careful.

1.1 Extrude Gear

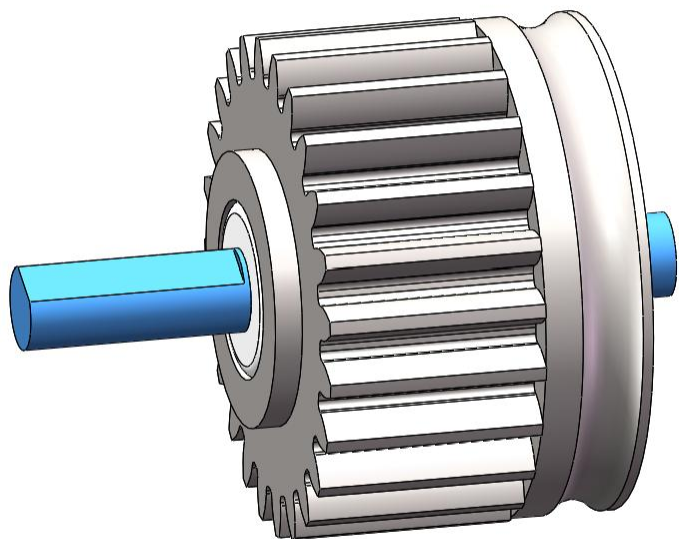


Extrude gear

You need four parts: an anti-slip one-way bearing, a nylon extrude gear, a metal extrude wheel, and a needle bearing. First, fit the nylon extrude gear with the metal extrude wheel, then press in the one-way bearing. Pay attention to the direction of the one-way bearing – one side has text and the other does not. One extrude gear needs the text side pressed in, the other needs the non-text side pressed in. Finally, install the needle bearing.



1.2 Side Pull Pinion

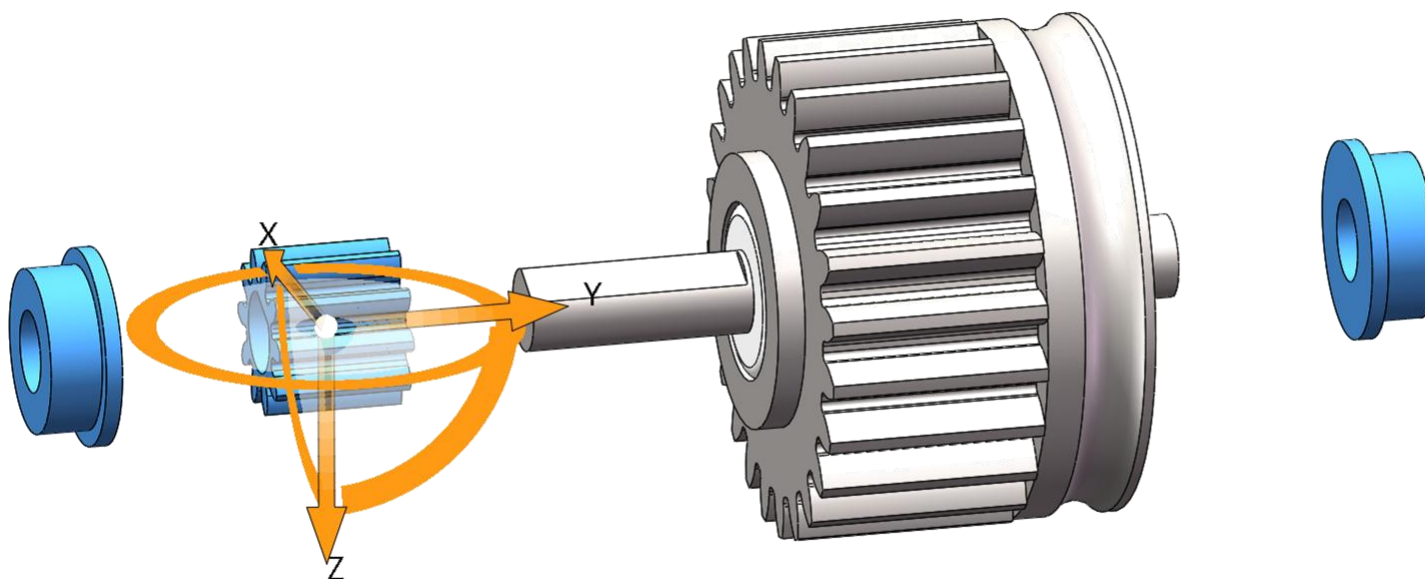


3mm gear axle

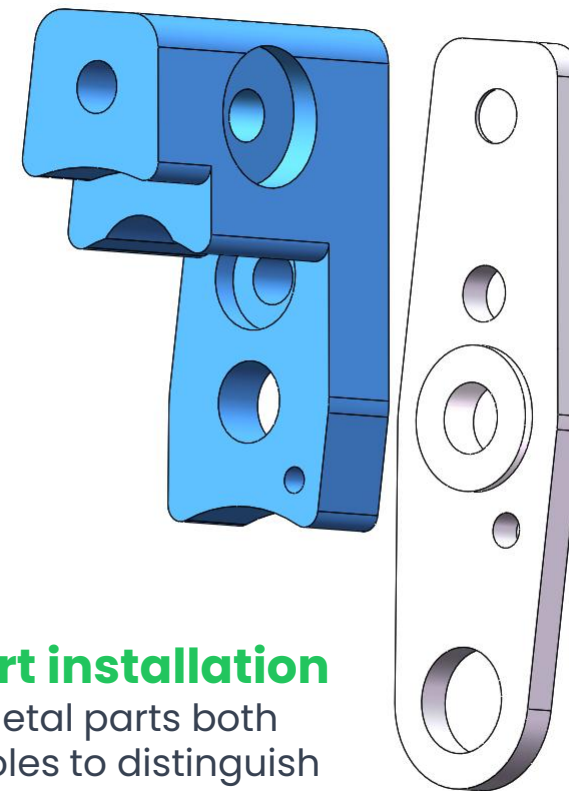
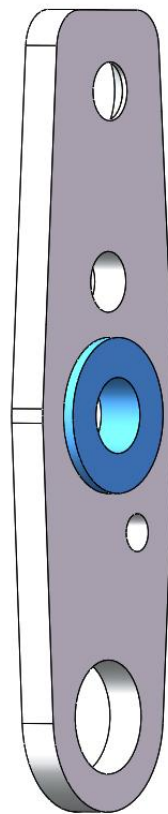
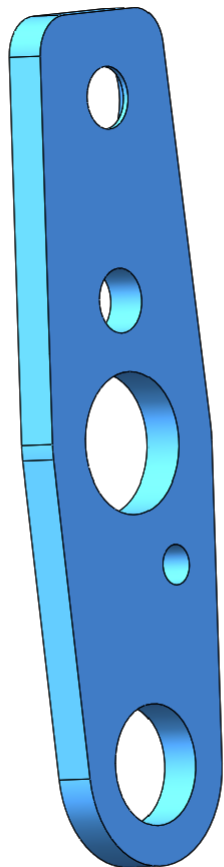
Install as shown.

0.5m10gear gear, 3mm flange bearing

Align the pinion to the D port and install it, then put two flange bearings on both sides.



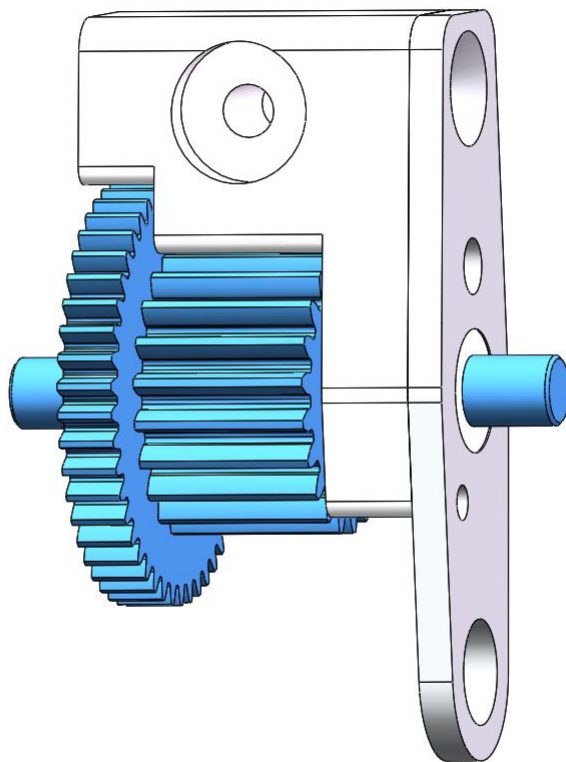
1.3 Gear Lever Part



Gear lever part installation

The plastic and metal parts both have two small holes to distinguish front from back. The small holes need to be on the same side for correct installation. Before securing the plastic part, remember to install a flange bearing with inner diameter 4 and outer diameter 8.

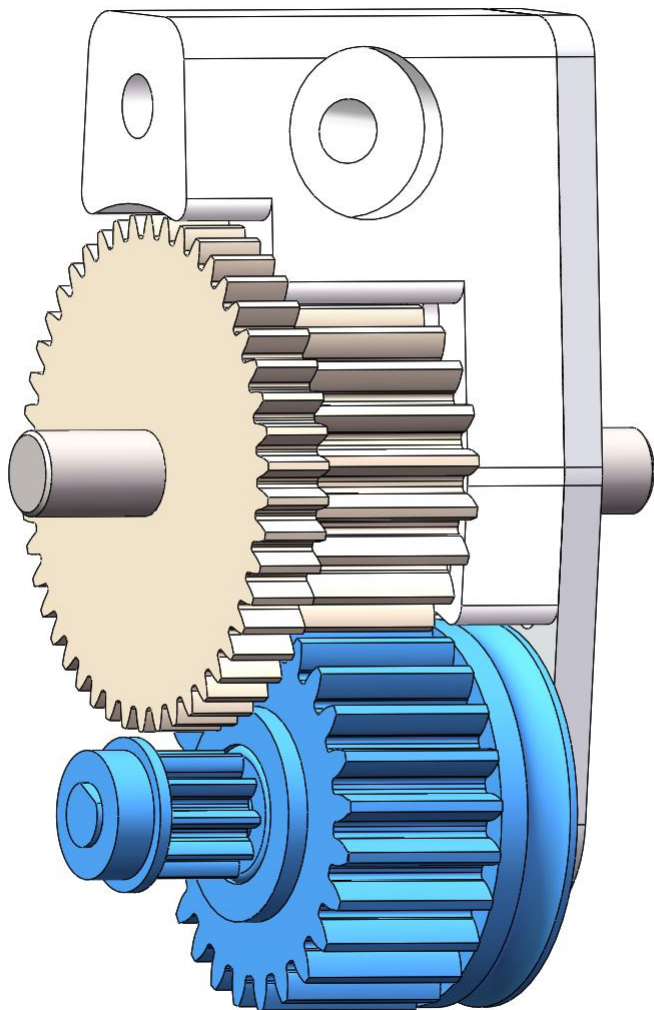
1.4 Double Layer Gear



Double layer gear, 4mm axle installation

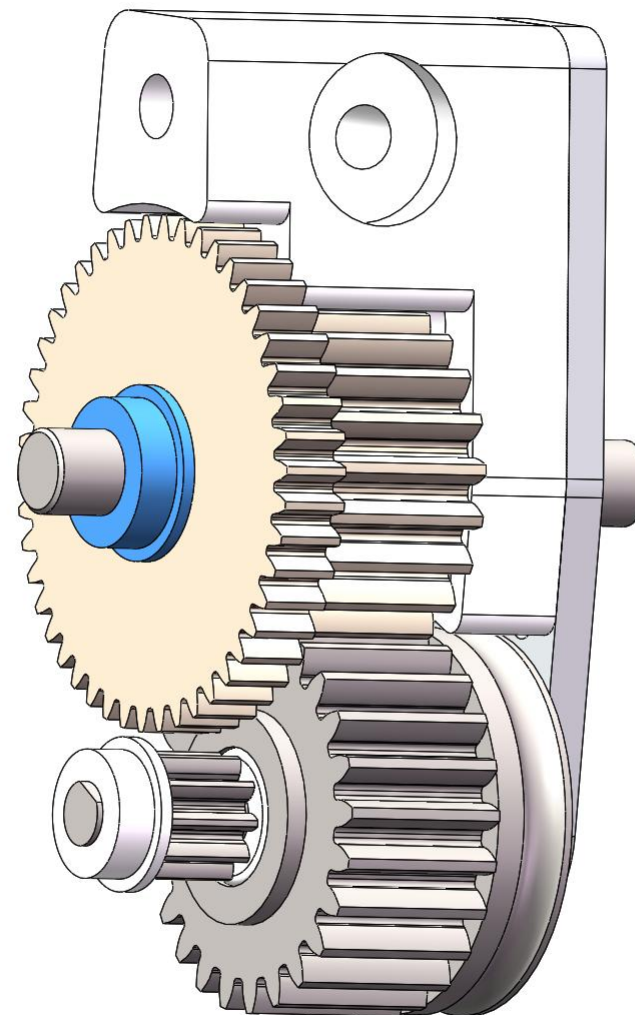
Install as shown.

1.5 Install the Extrude Gear Set

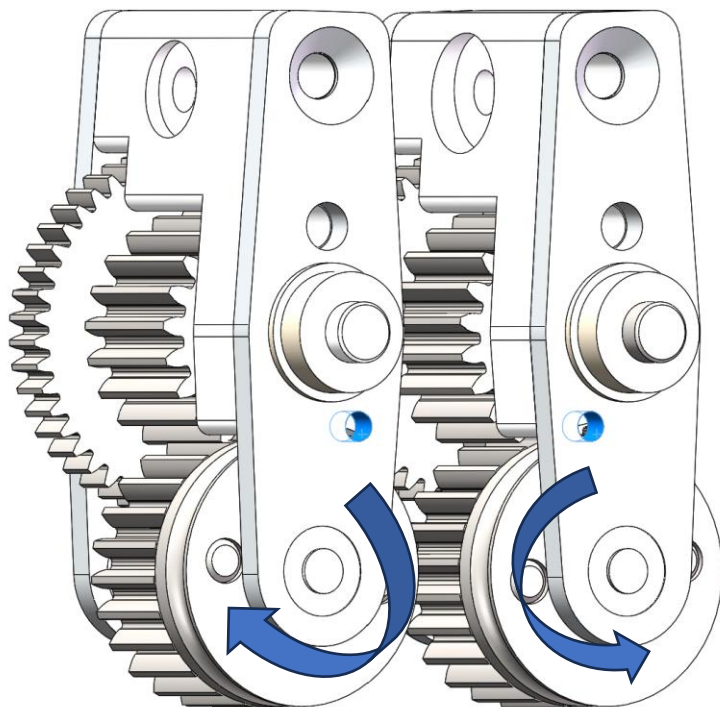


Extrude gear set installation

Install the extrude gear set as shown, and install the flange bearing with inner diameter 4 and outer diameter 7. Please see the next page for the gear set installation direction.



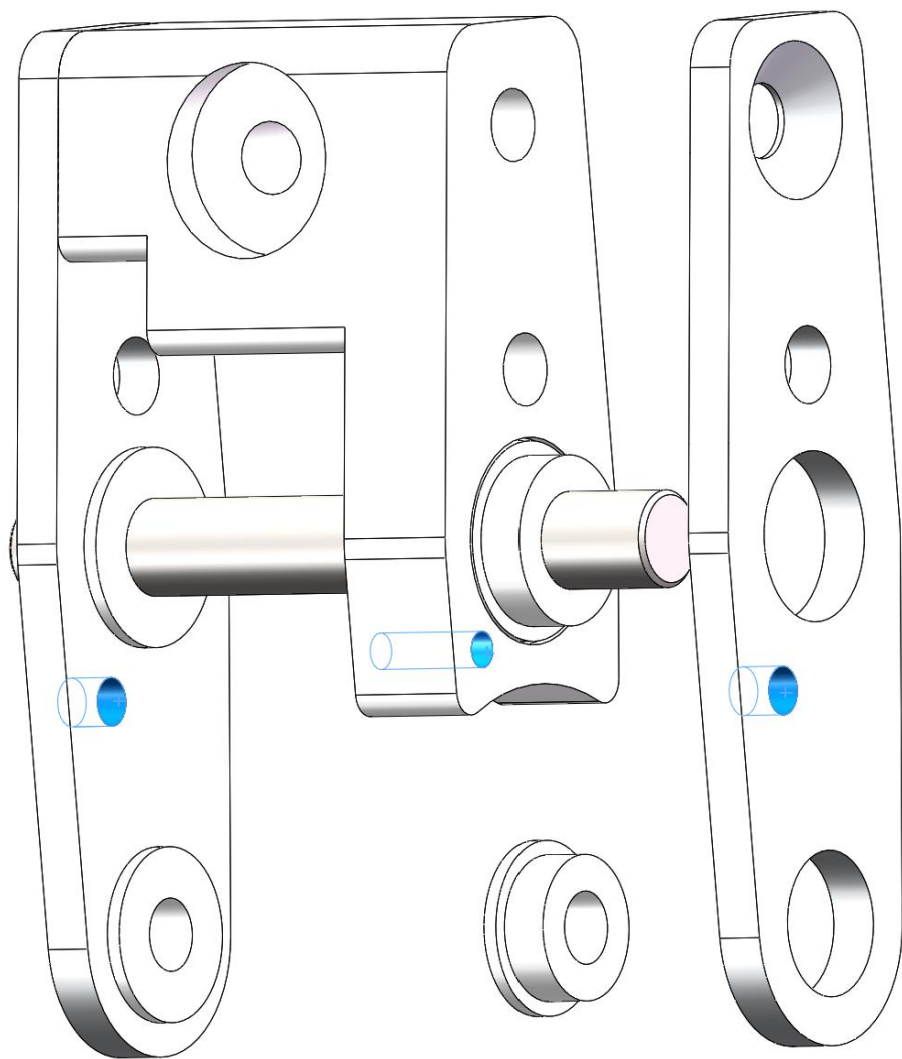
1.5 Install the Extrude Gear Set



Extrude gear set direction

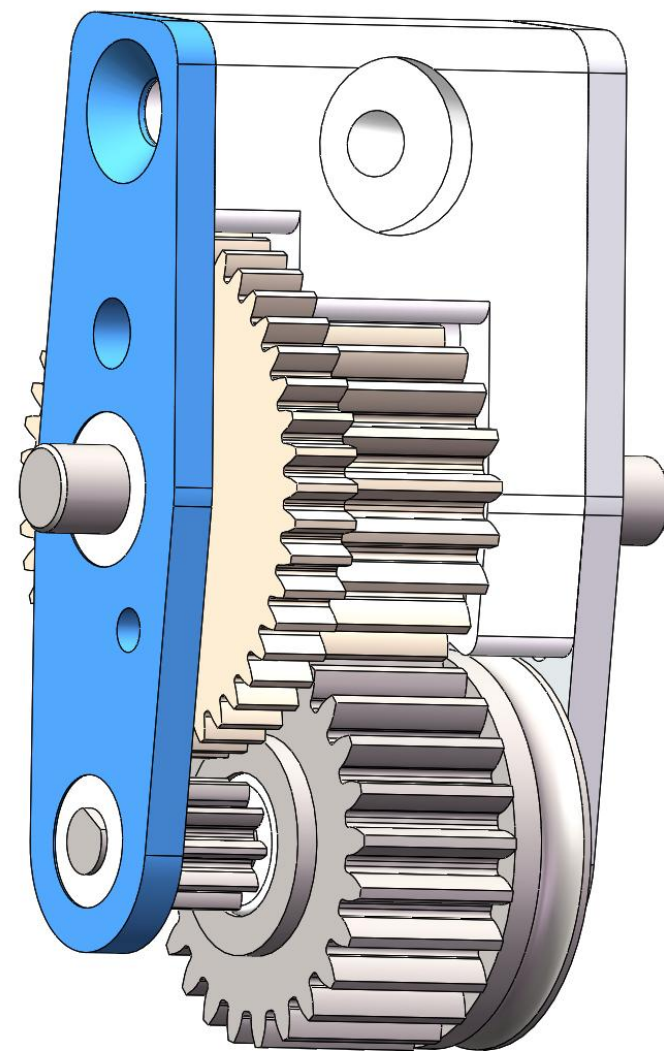
These are two mirrored extrude gear sets. Pinch the small pinion below with your hand and rotate the extrude wheel above. The extrude wheel rotating smoothly in the direction shown in the diagram means the direction is correct. The two positioning holes are on the inside.

1.6 Install the Mirrored Gear Lever Part

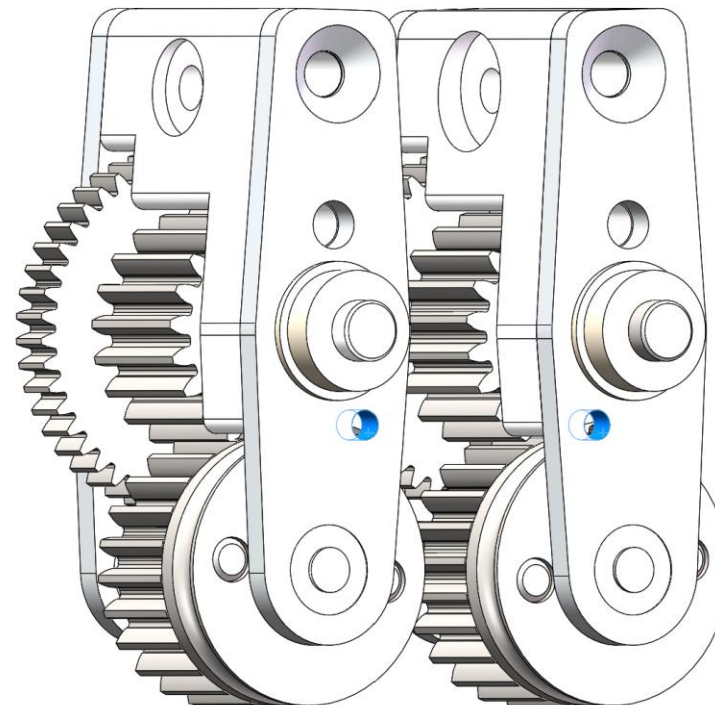
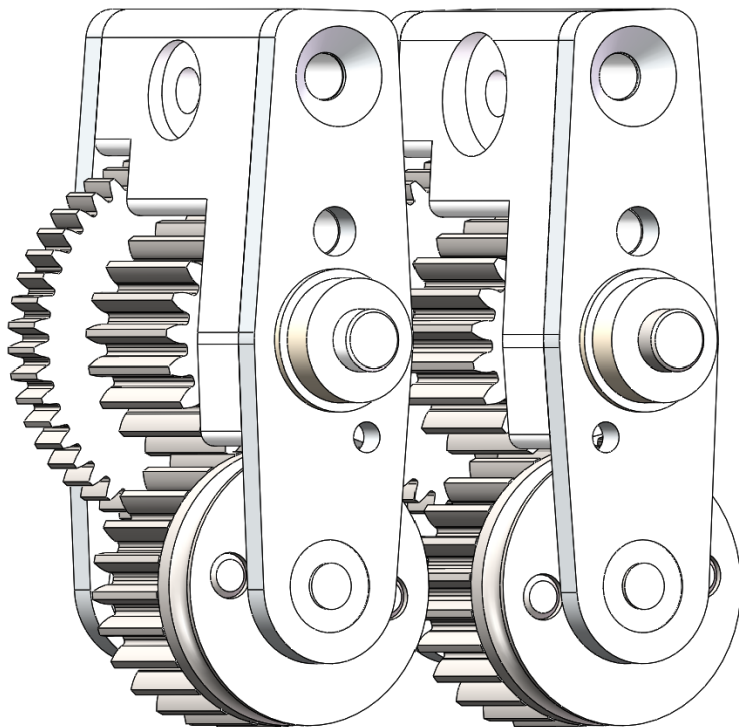


Gear lever part installation

Select the other gear lever part that is the mirror of the existing one. Note that the three positioning holes are on one side. Install as shown.



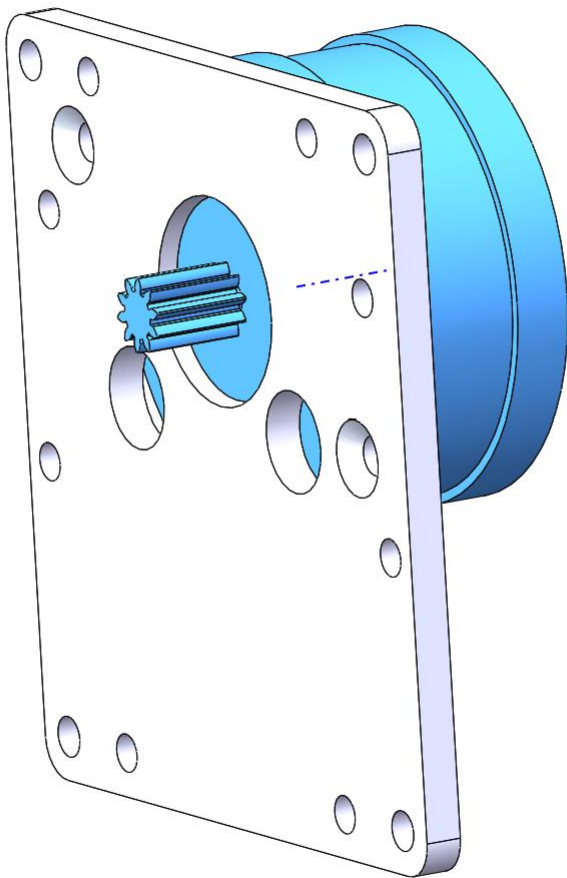
1.7 The Two Mirrored Gear Sets



Two gear sets

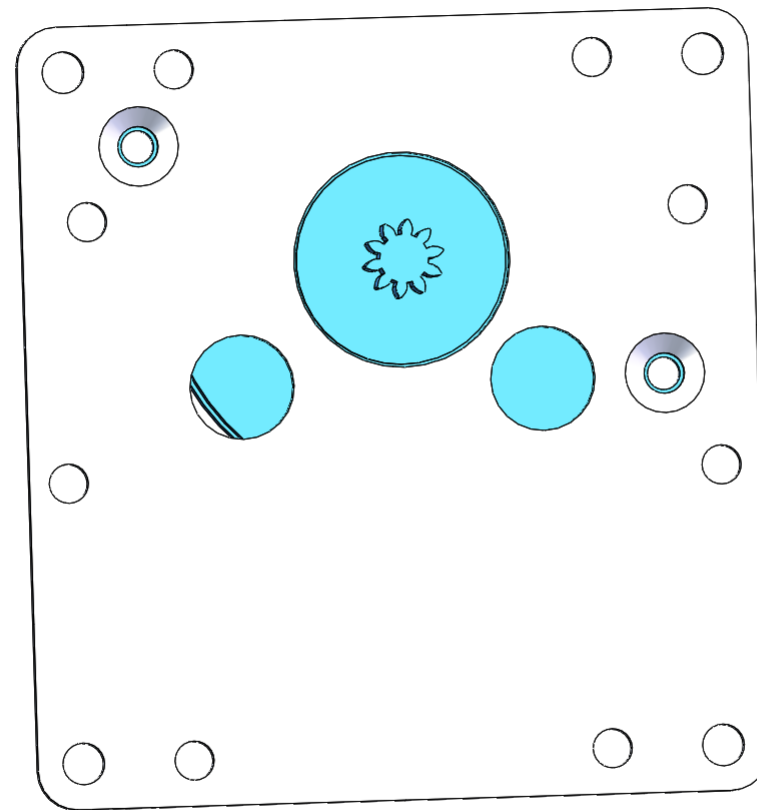
As shown, the two positioning holes facing inward is the correct direction.

02 Install the 36 Stepper Motor

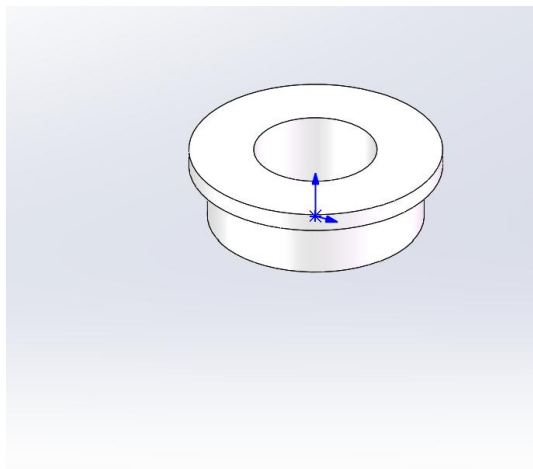


36 stepper motor, extruder bottom metal part

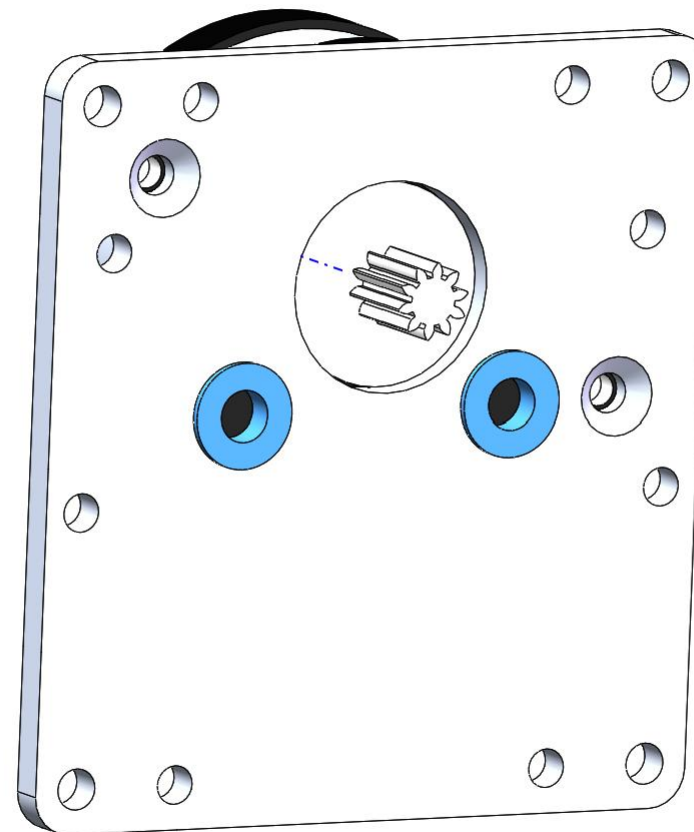
As shown, use M3*5 countersunk screws. It is recommended to install with the motor wiring facing down.



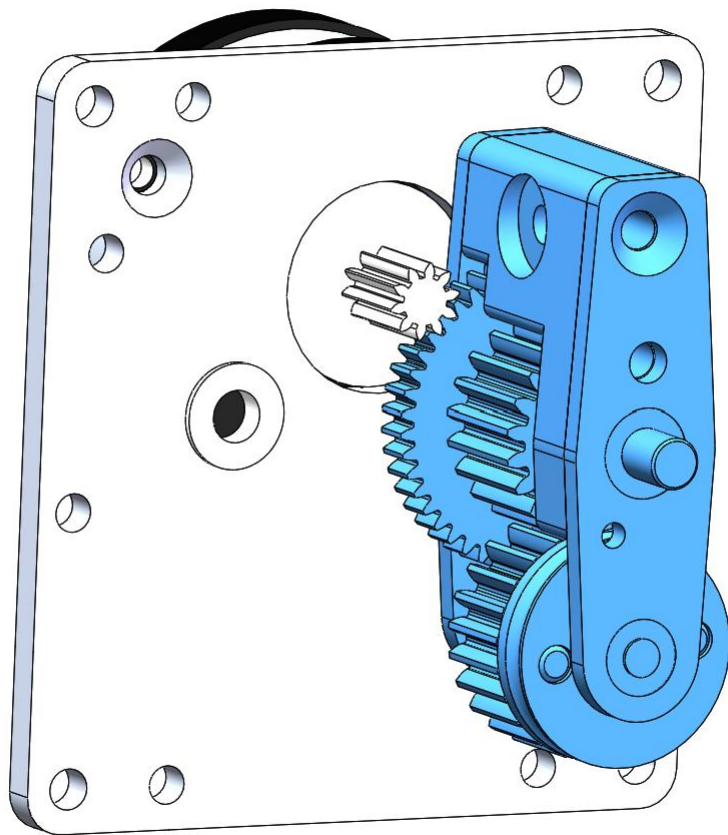
03 Install the Flange Bearings



4mm flange bearing (OD 7)
Install as shown.

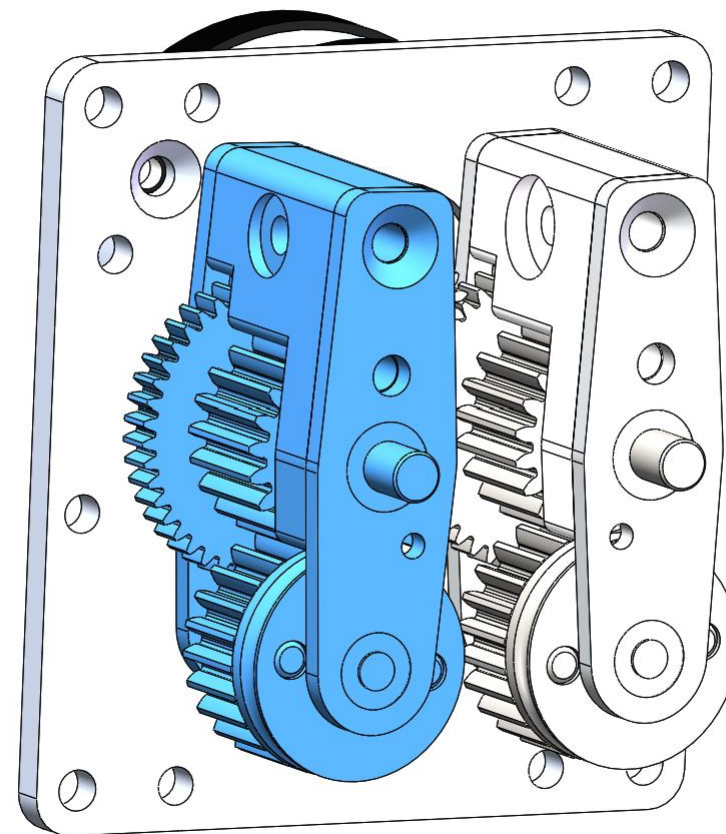


04 Install the Gear Set

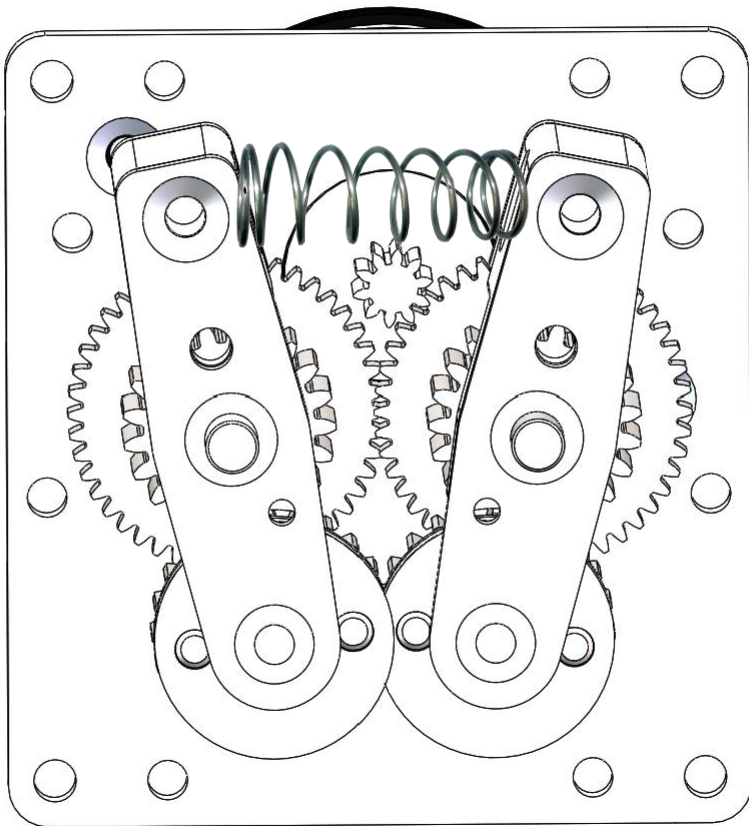


Two metamorphic gear sets

As shown, the two positioning holes facing inward is the correct direction.

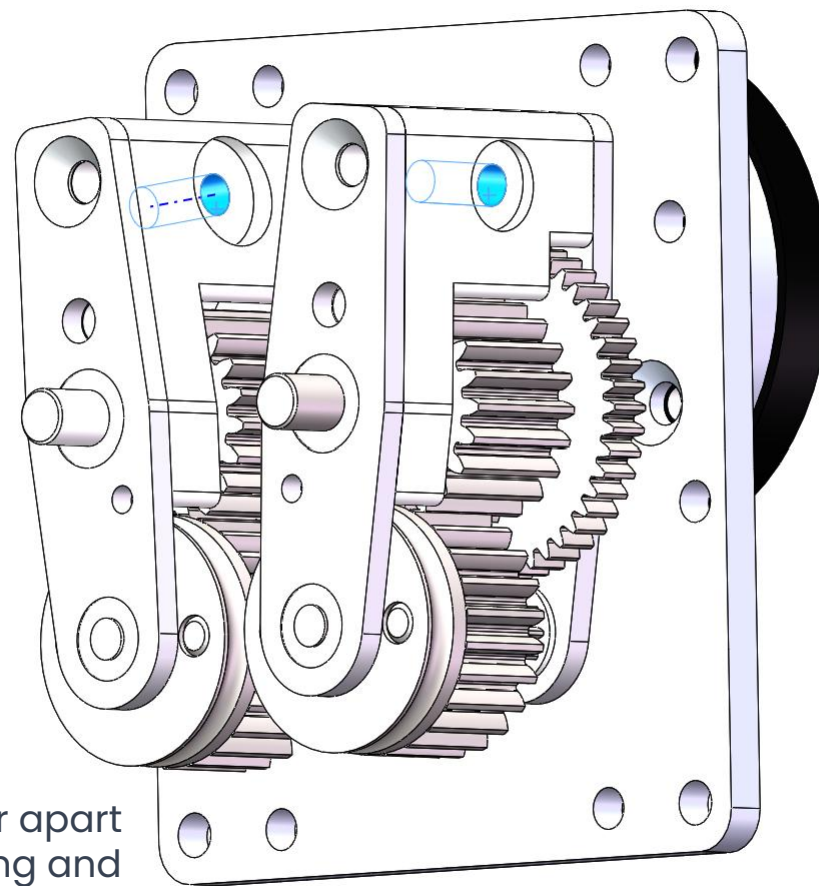


05 Install the Spring

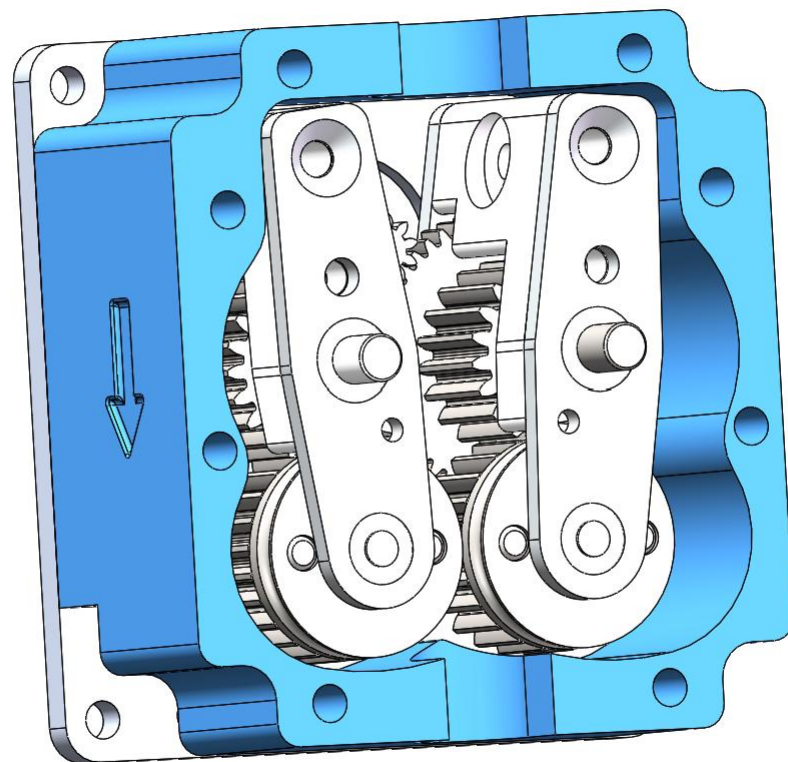


Spring

Spread the two gear sets as far apart as possible, compress the spring and install it into the slot, then screw 2 M3 screws on the left and right sides to prevent the spring from coming loose. (Spring was drawn by hand since it was not modeled in SolidWorks.)



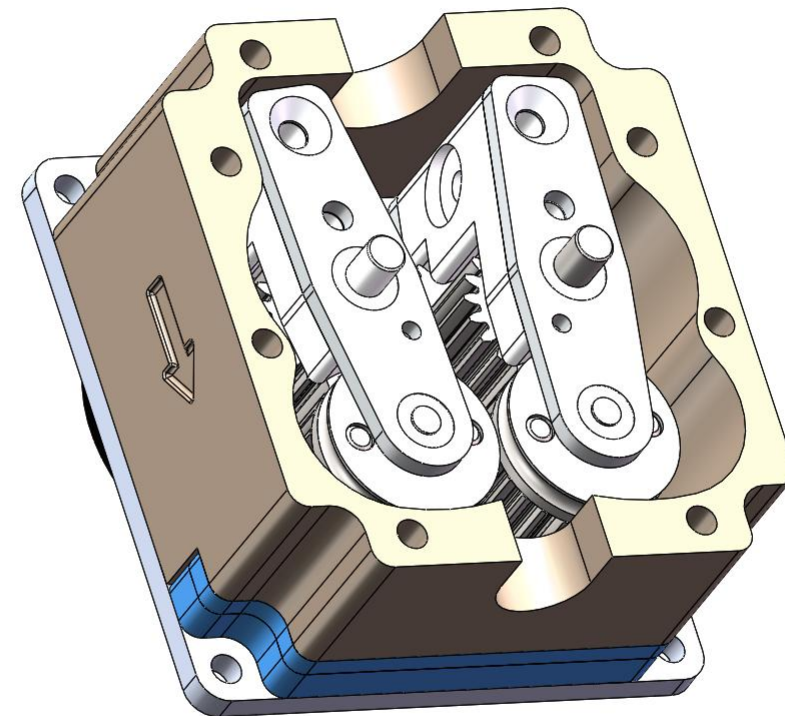
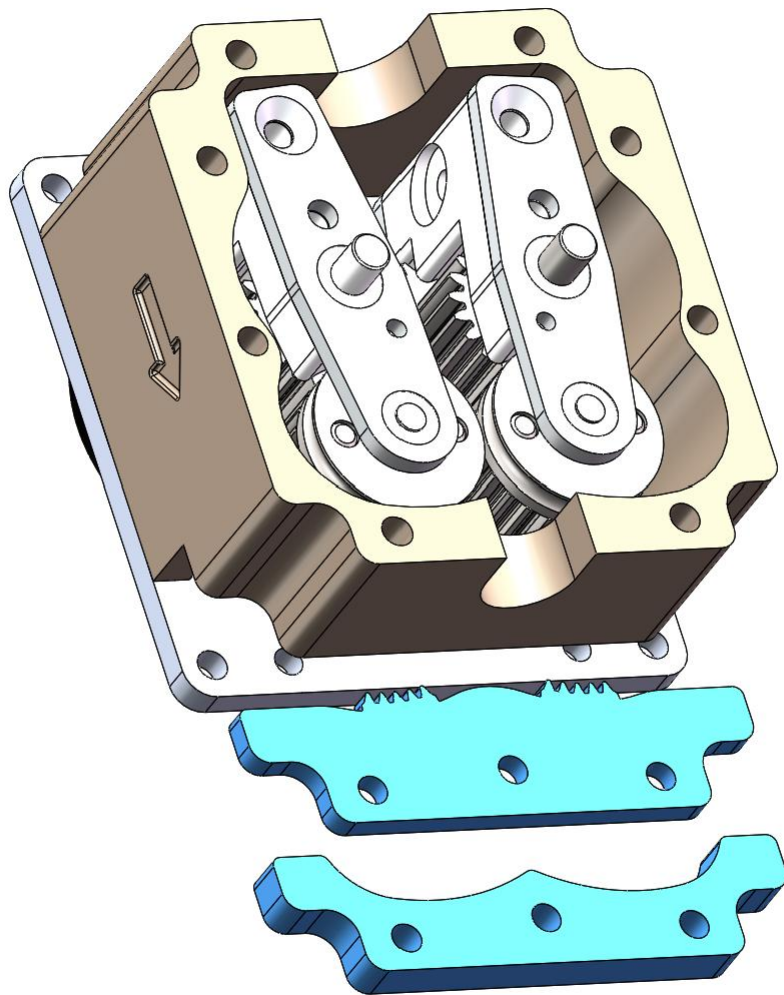
06 Install the Extruder Outer Shell



Extruder outer shell (printed part)

Compress the spring, then
slide the outer shell on.

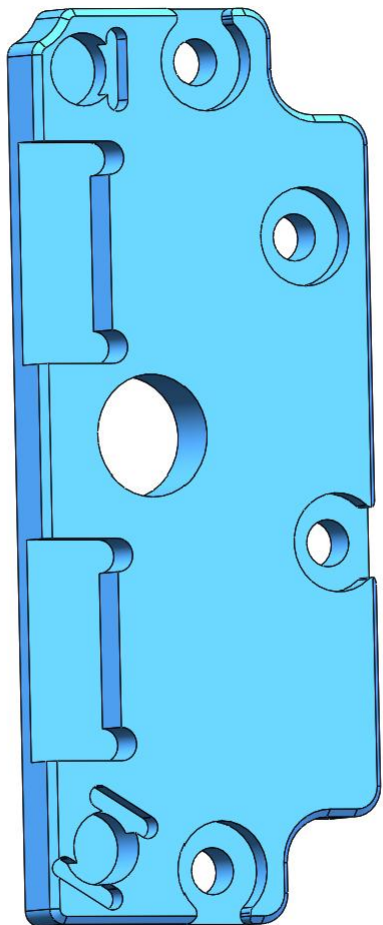
07 Install the Wire-cut Rack



Wire-cut rack, rack below part

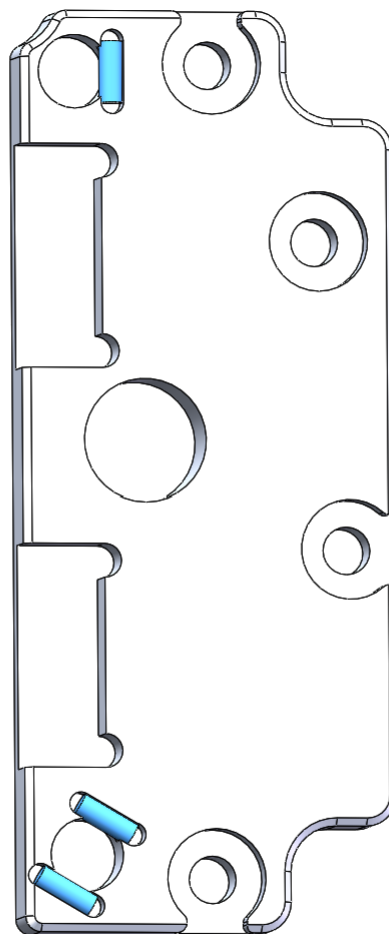
As shown, install the rack on top and the raiser below. If the rack is not seated correctly and looks crooked, rotate the extrude wheel a couple of turns.

8.1 Glue the Positioning Pins and Magnets



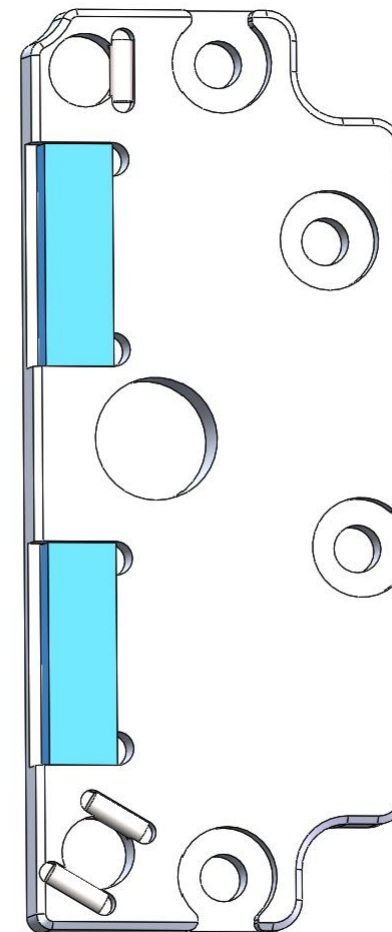
Extruder top cover, 1.5*4 cylinder pin

As shown, glue the cylinder pins.

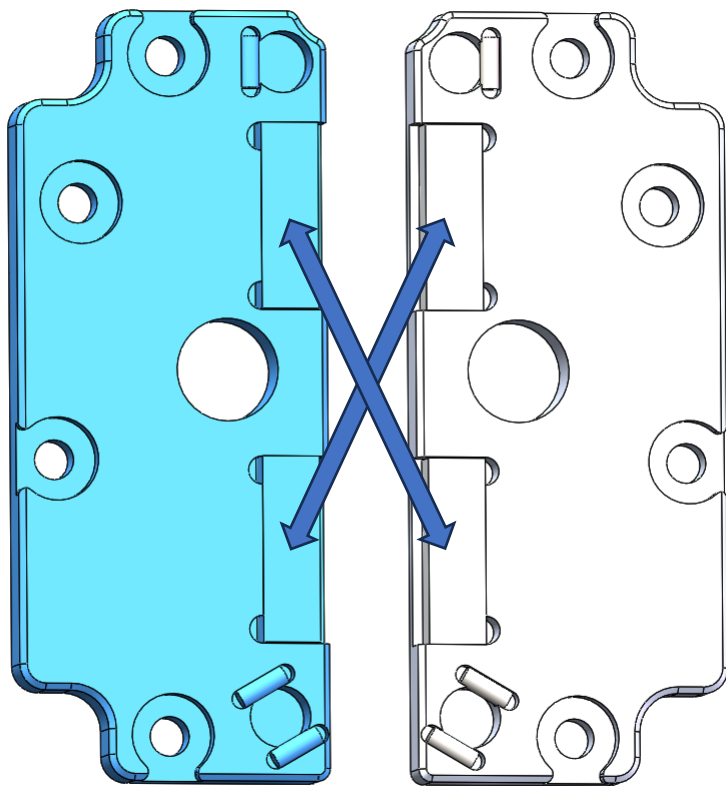


Magnet 15*5*3

As shown, glue the magnets. It is recommended to glue the two magnets in opposite directions (more details on the next page).

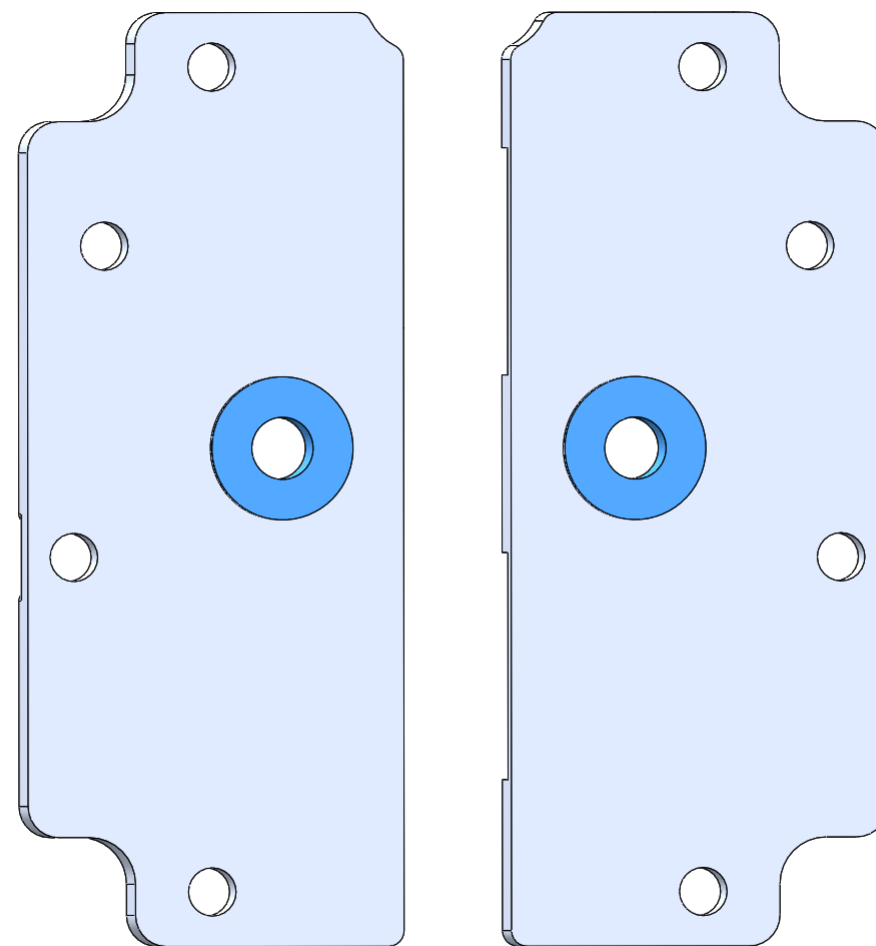


8.1 Glue the Magnets



Magnet 15*5*3

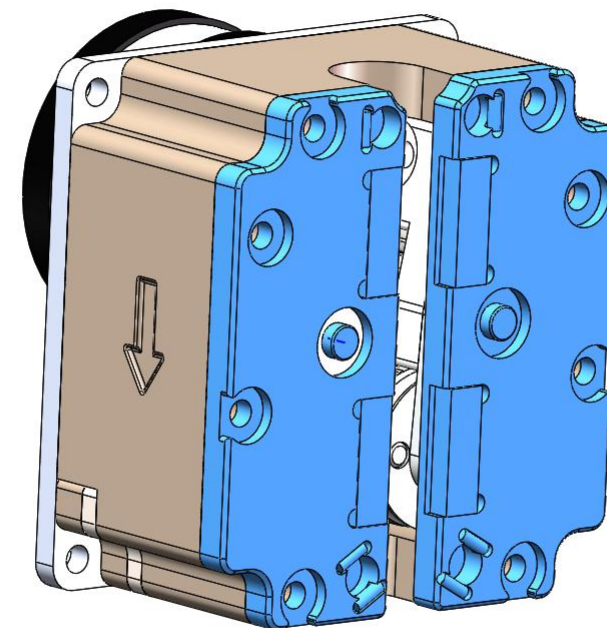
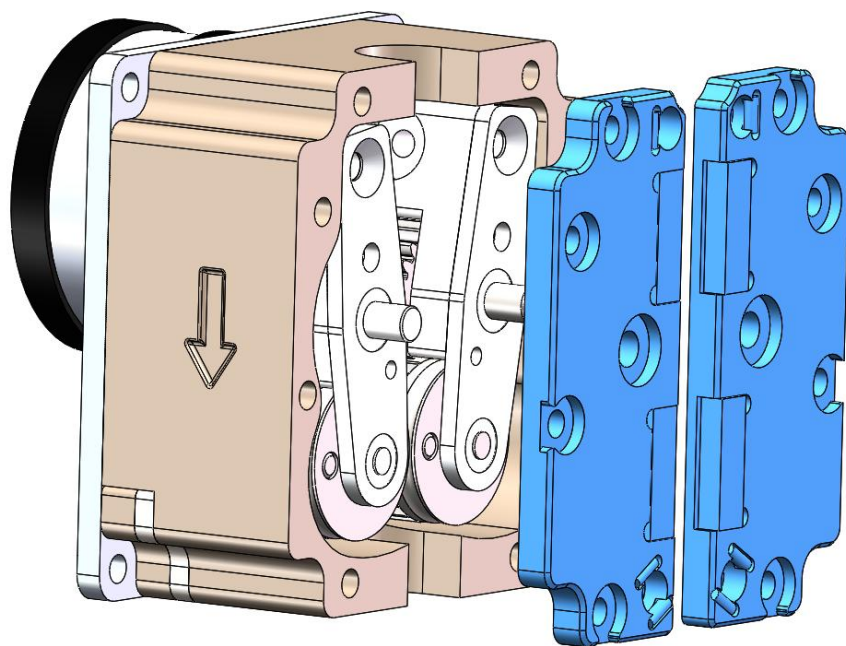
As shown, glue the magnets. Top-left and bottom-right are the same magnetic pole, and top-right and bottom-left are also the same pole. Gluing the magnets this way makes the magnetic field lines more closed-loop, giving relatively better magnetic force and making the magnets less likely to come loose.



4mm flange bearing (OD 8)

Install as shown.

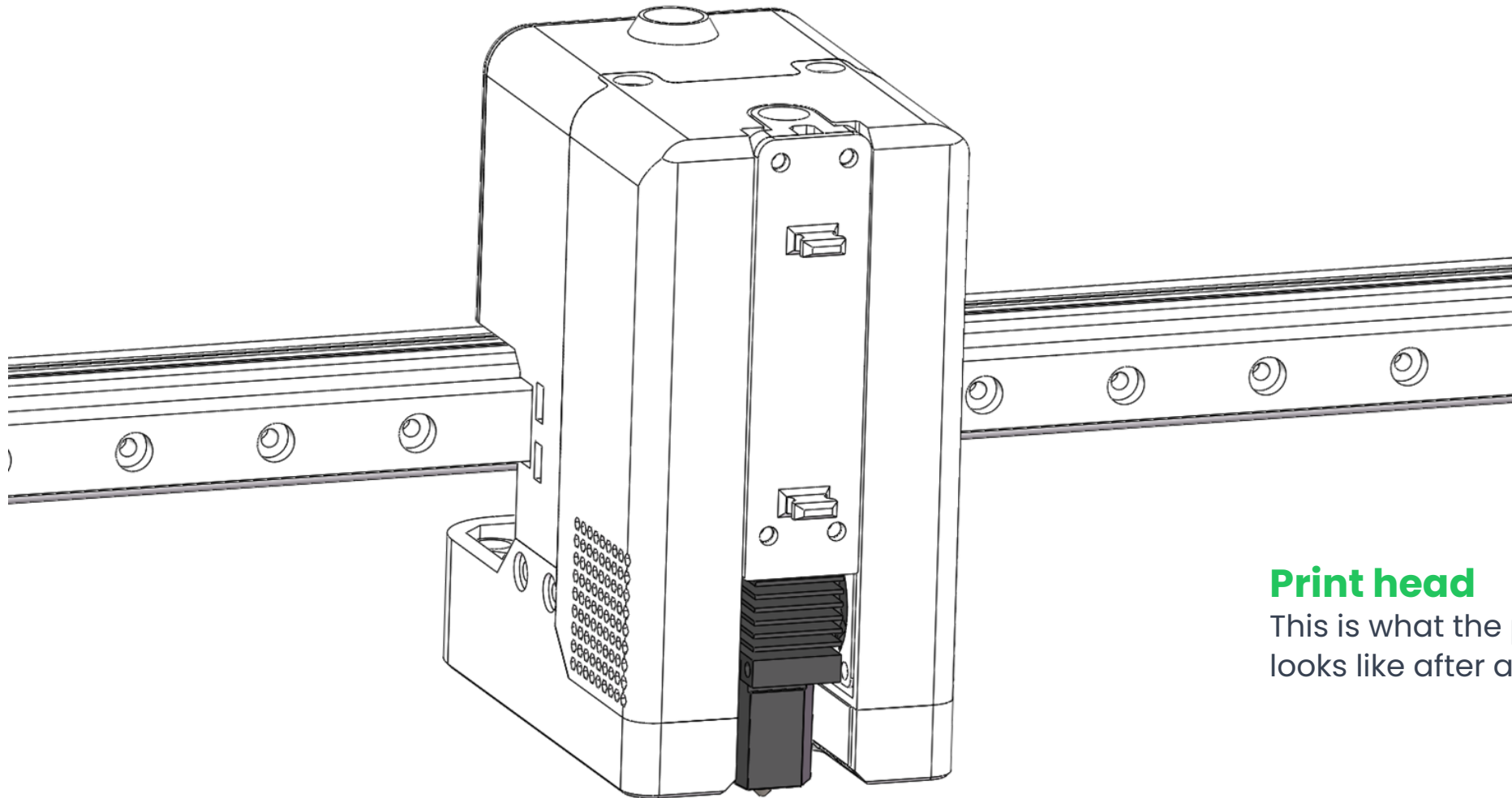
09 Install the Extruder Front Cover



M3*30 screw

As shown, use 8 M3*30 screws to secure the extruder front cover.

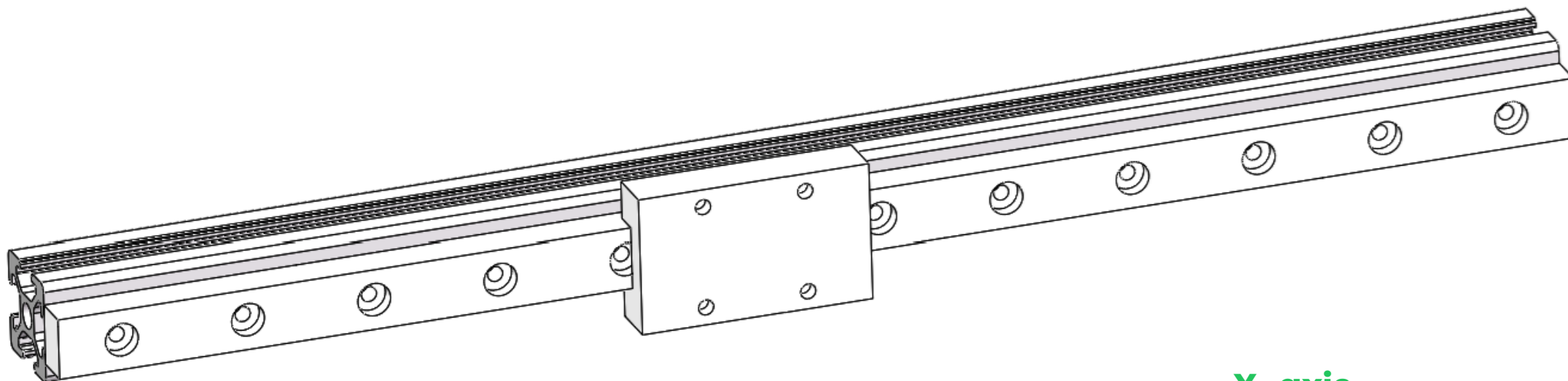
Part 2: Print Head Assembly



Print head

This is what the print head looks like after assembly.

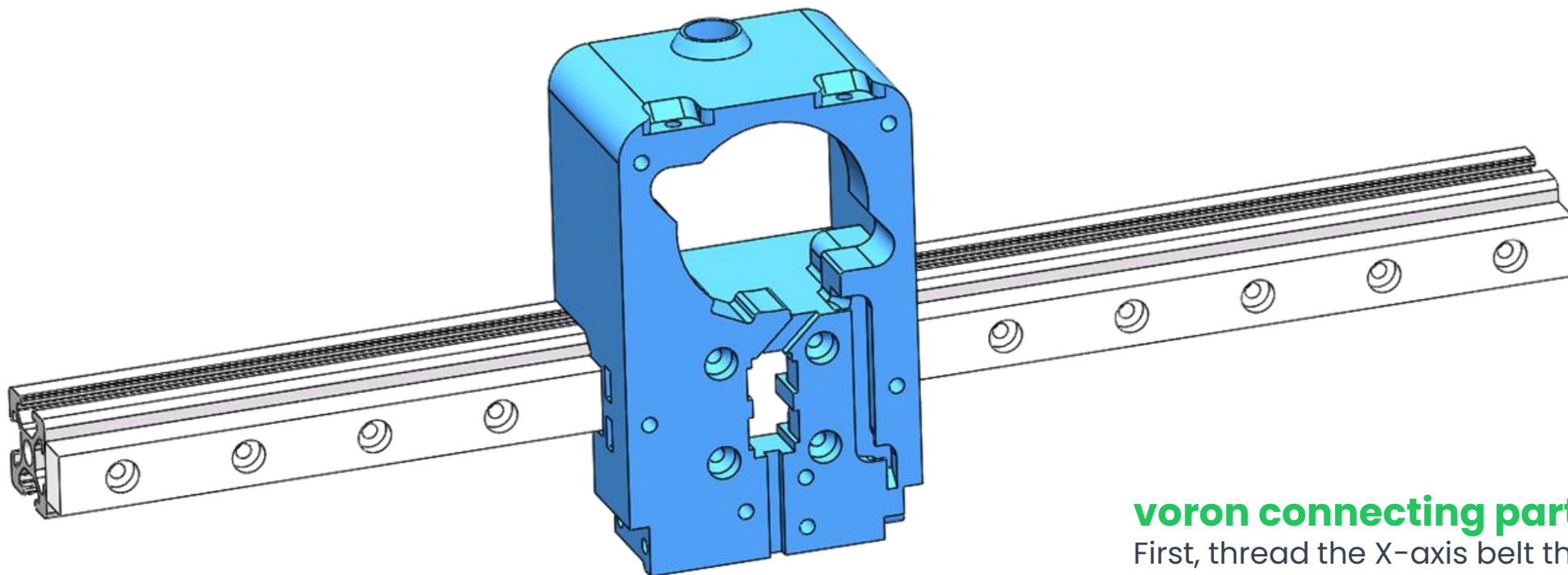
01 X-axis



X-axis

This is the X-axis of a VORON printer

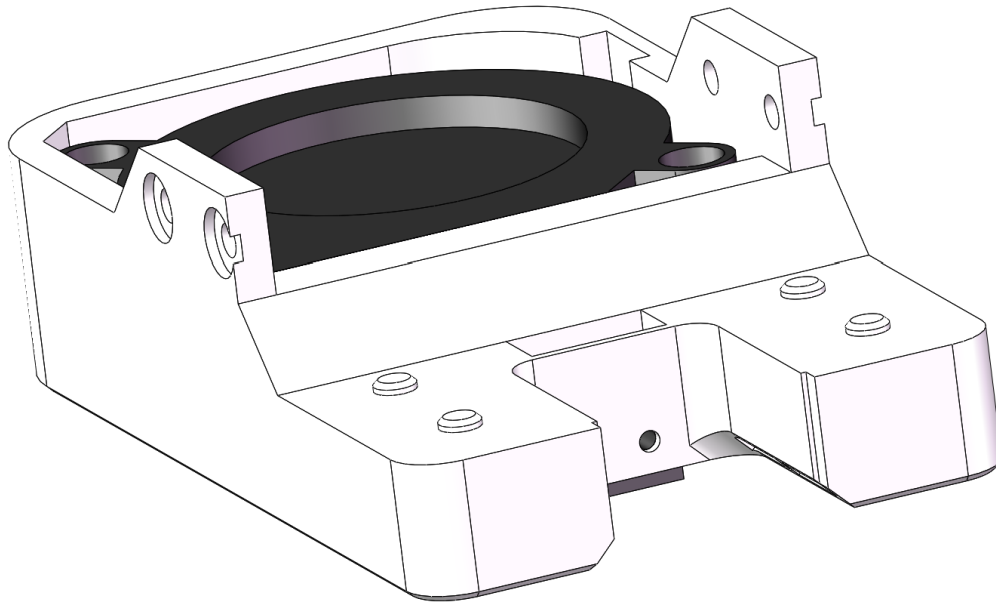
02 Install the Structural Mounting Parts



voron connecting part

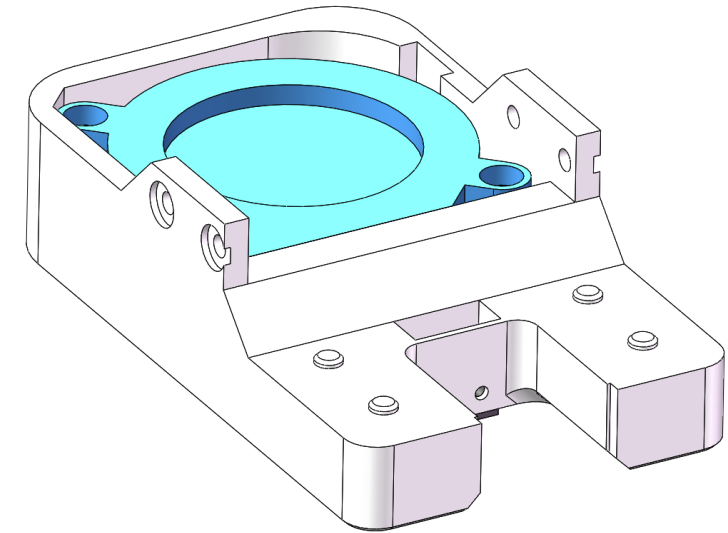
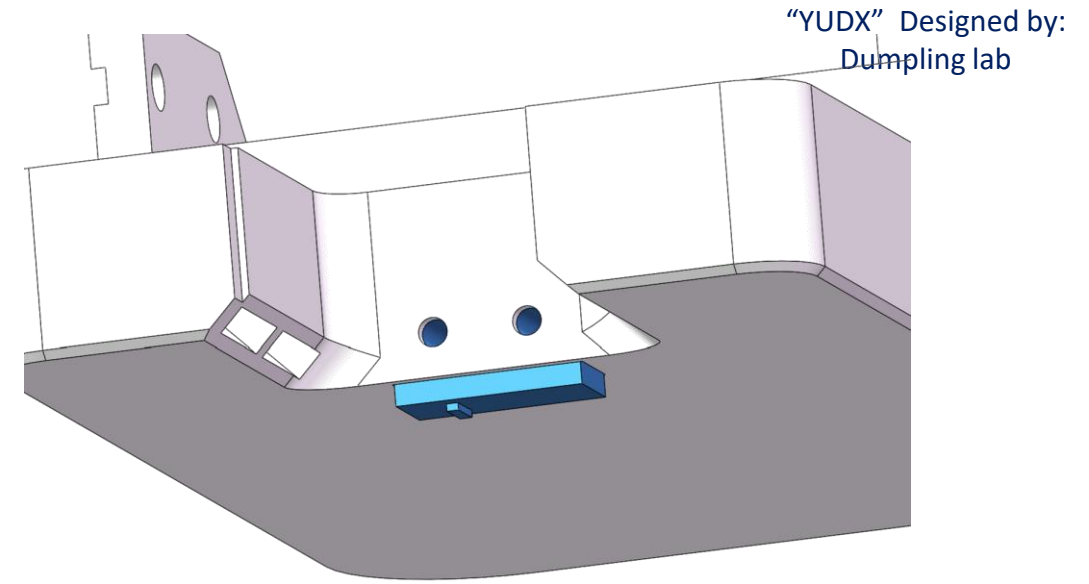
First, thread the X-axis belt through the belt hole on the side to the front, the same way the belt is installed on the VORON 2.4 SB print head. Then use screws to secure the print head.

3.1 Model Cooling Fan Section

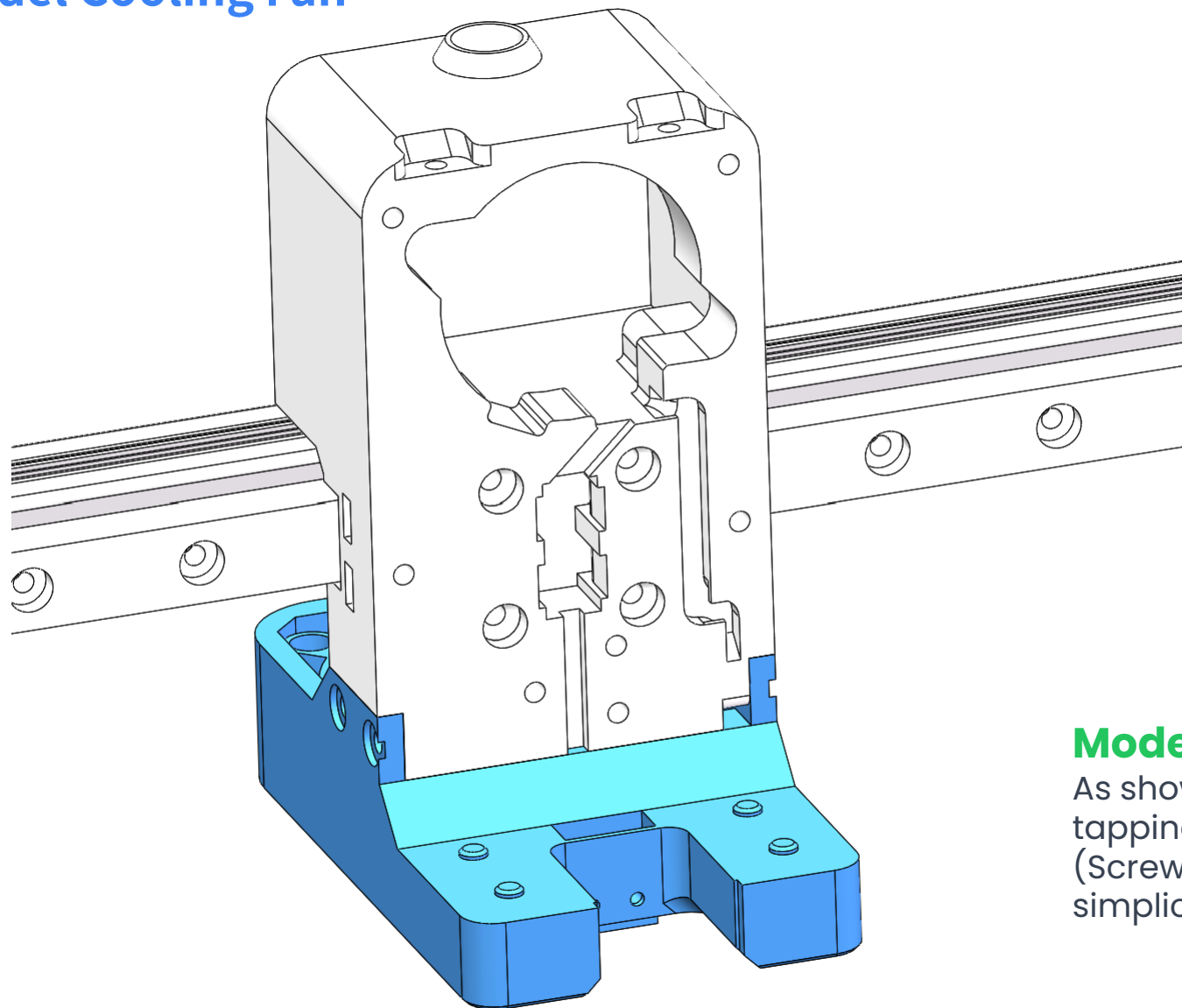


5020fan, bed switch

Secure the 5020 fan and micro switch as shown. Remember to solder the wires on the micro switch.



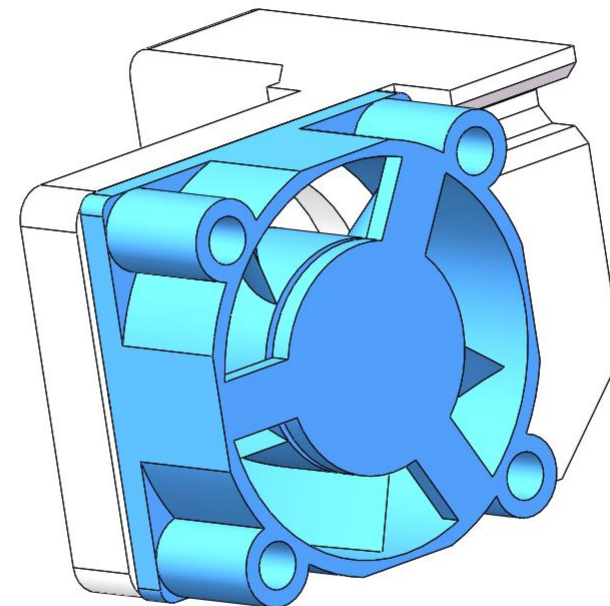
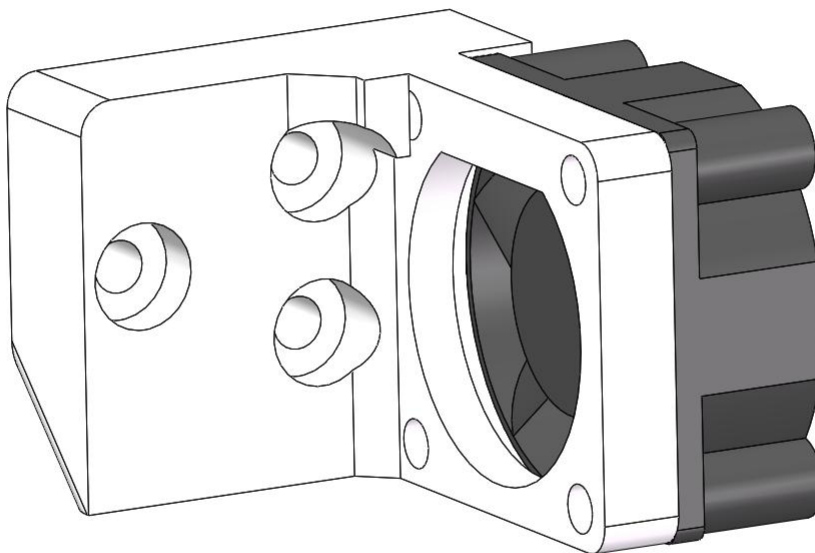
03 Secure the Model Cooling Fan



Model cooling fan

As shown, secure with 4 self-tapping screws from the side. (Screws not drawn for simplicity.)

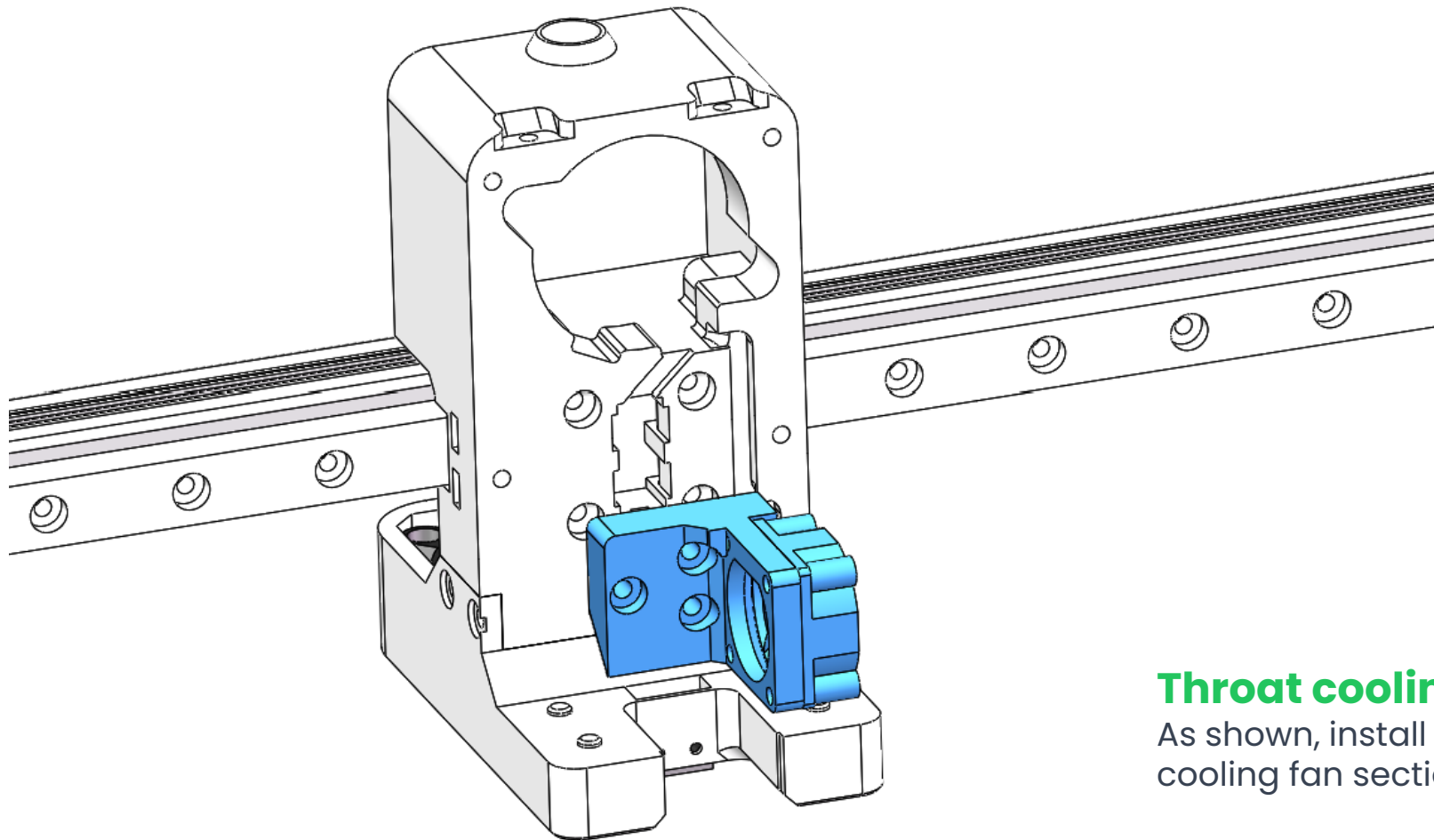
4.1 Throat Cooling Fan Section



2510fan, throat cooling fan seat

As shown, secure the 2510 fan.
Route the fan wires through the notch in the printed part.

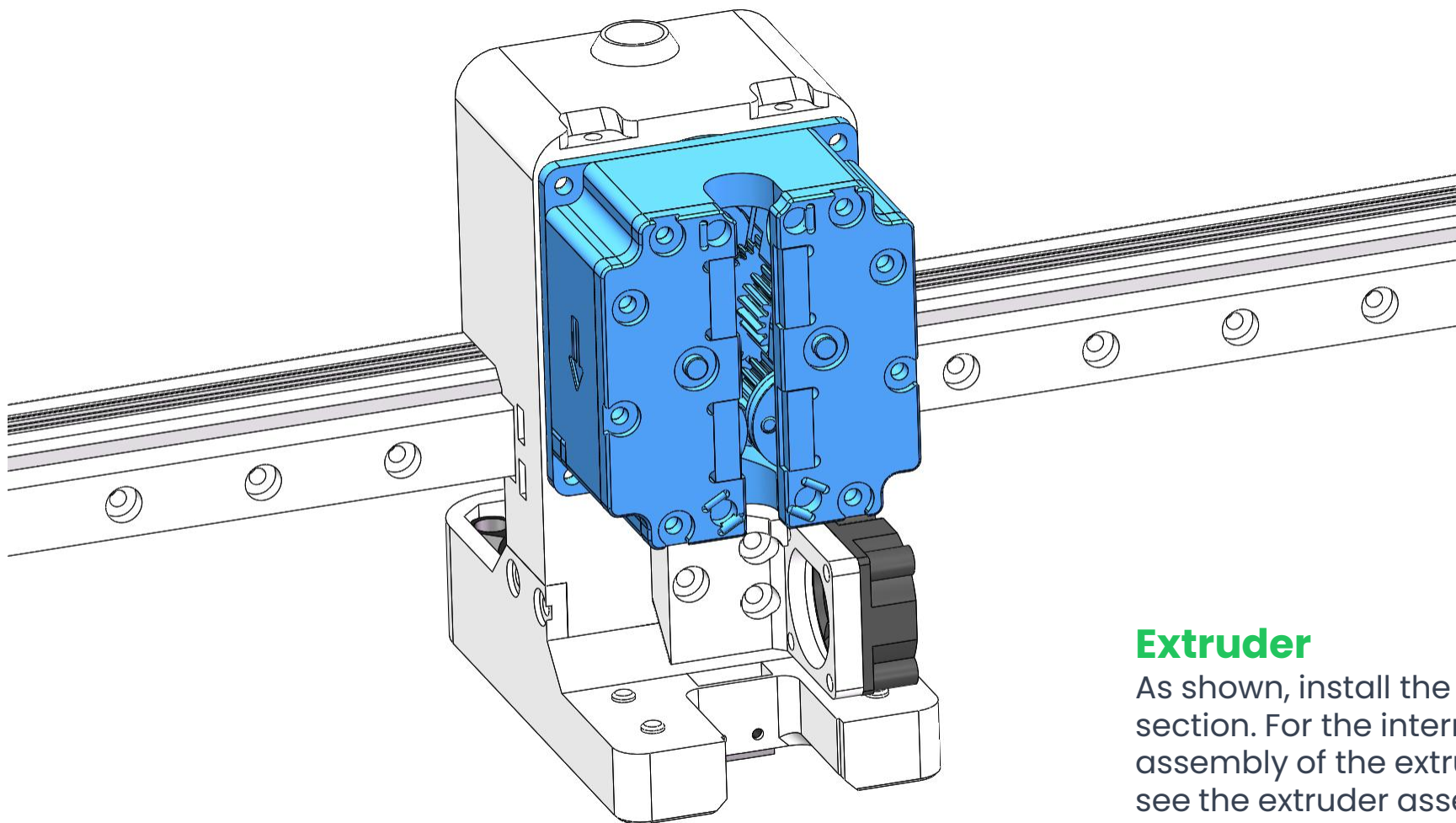
04 Secure the Throat Cooling Fan



Throat cooling fan

As shown, install the throat cooling fan section.

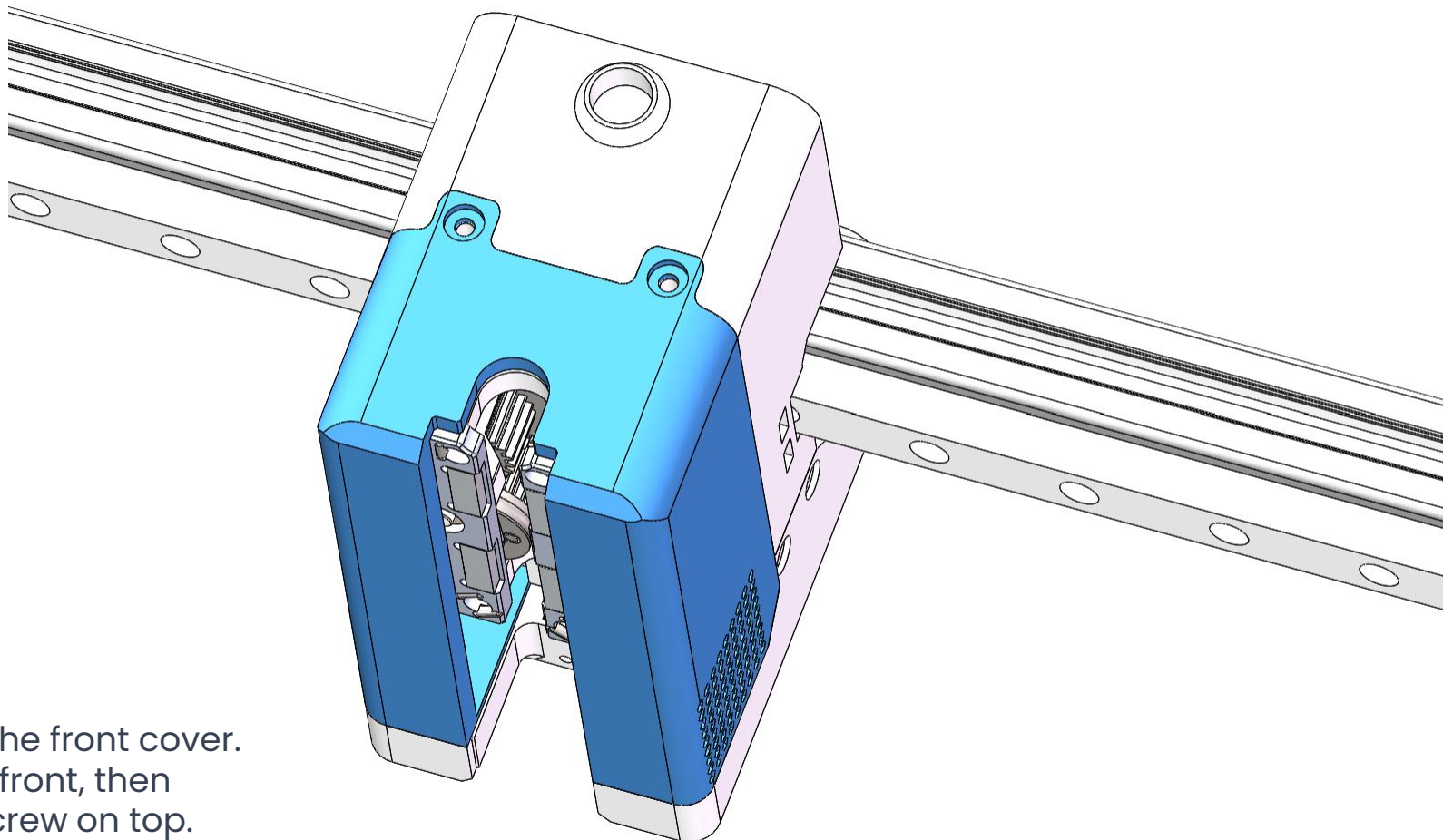
05 Install the Extruder



Extruder

As shown, install the extruder section. For the internal assembly of the extruder, please see the extruder assembly tutorial.

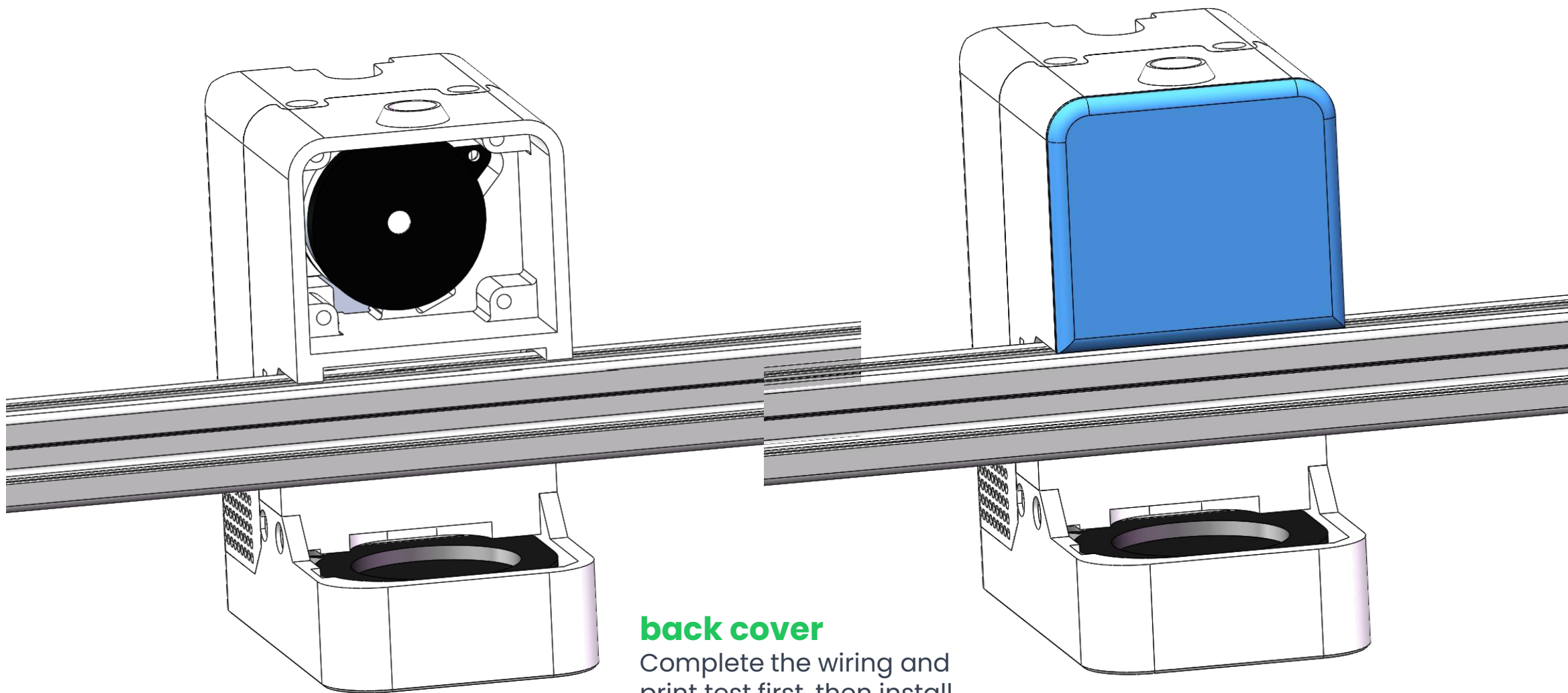
06 Install the Front Cover



front cover

As shown, install the front cover.
Clip it in from the front, then
secure with the screw on top.

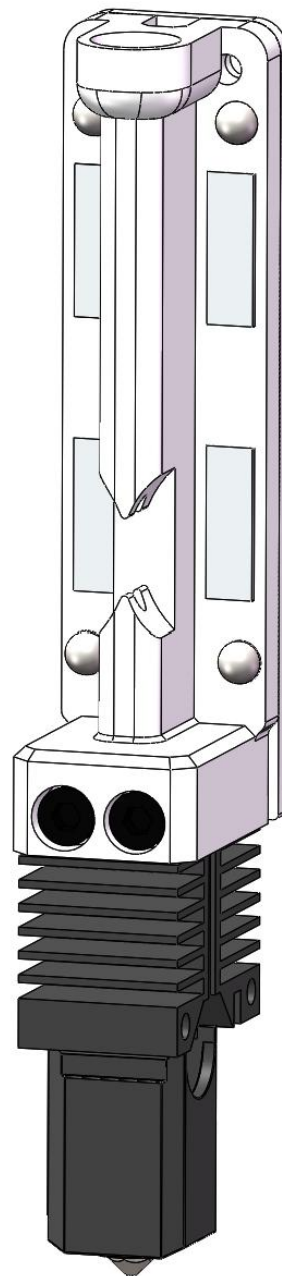
07 Install the Back Cover



back cover

Complete the wiring and
print test first, then install
the back cover last.

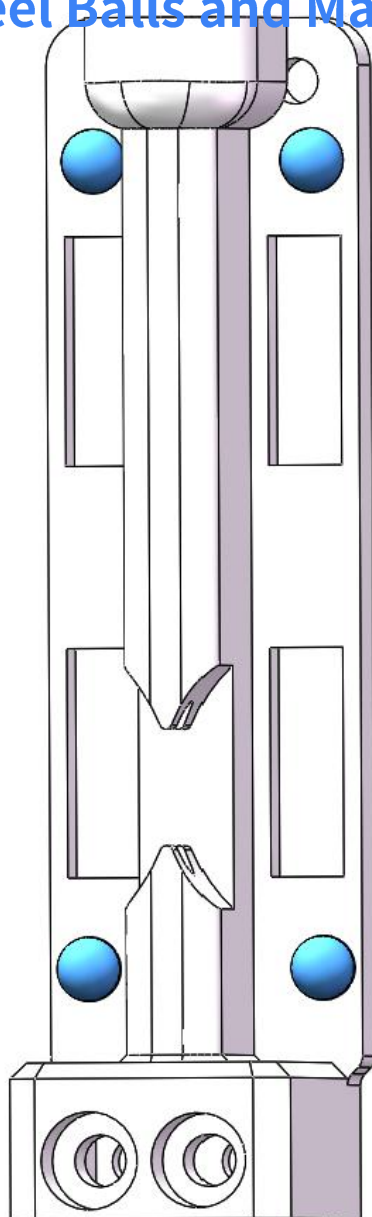
Part 3: Tool Head Assem



Tool head

This is what the tool head looks like after assembly.

01 Glue the Steel Balls and Magnets

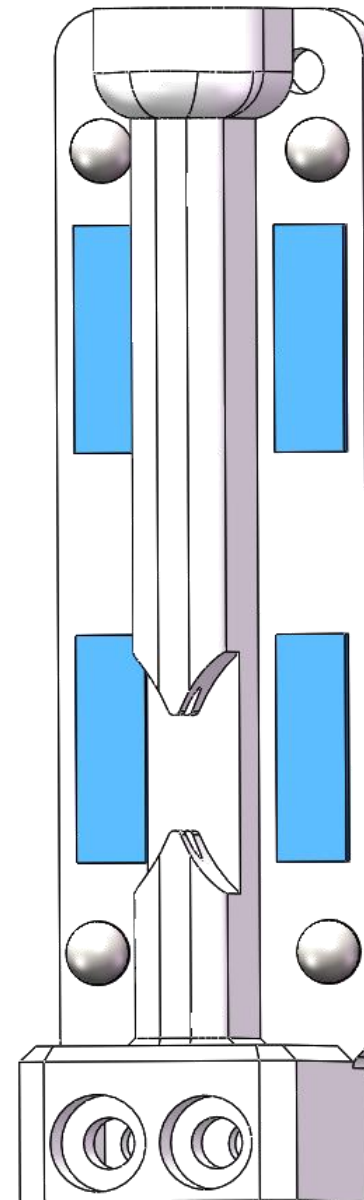


4mm steel ball

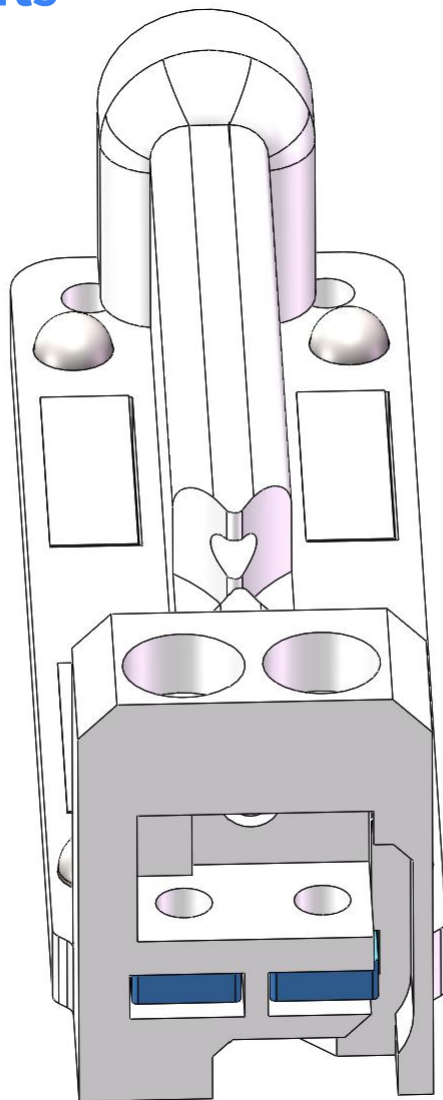
Put a drop of instant glue, place the steel ball in, and press it with a hard object to make sure it sticks to the bottom.

Magnet

When gluing, pay attention to the polarity of the extruder magnets. The diagonal corners are different magnetic poles, and should attract the extruder magnets.



02 Insert the Square Nuts

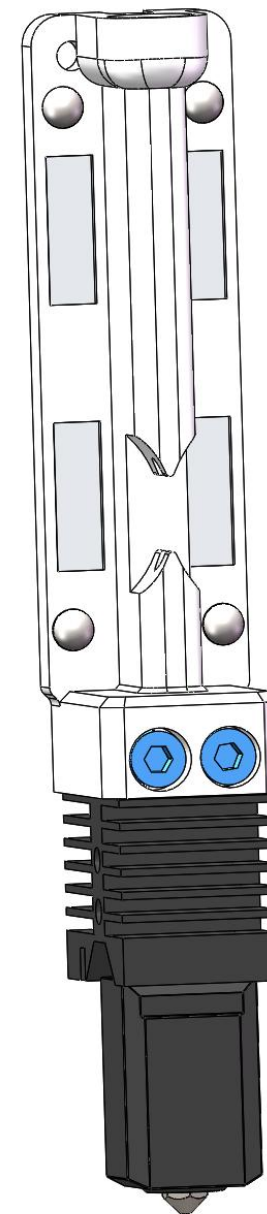
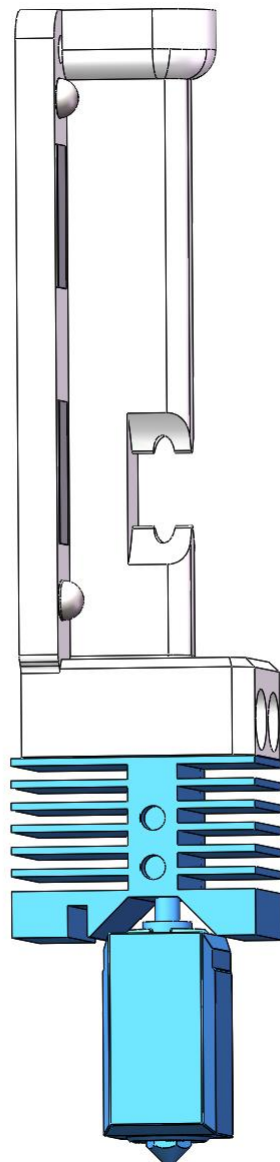


M3 square nut
Just drop them in.

03 Install the Bamboo Head

Bamboo head

Pay attention to the installation direction. Route the wires through the reserved wire routing slot.



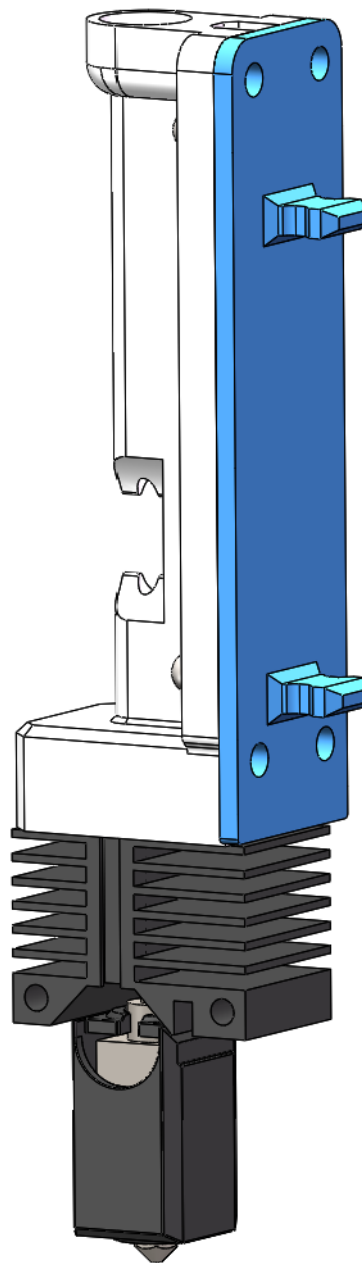
M3*14 sleeve screw

Secure as shown.

04 Install the Back Plate

Wiring

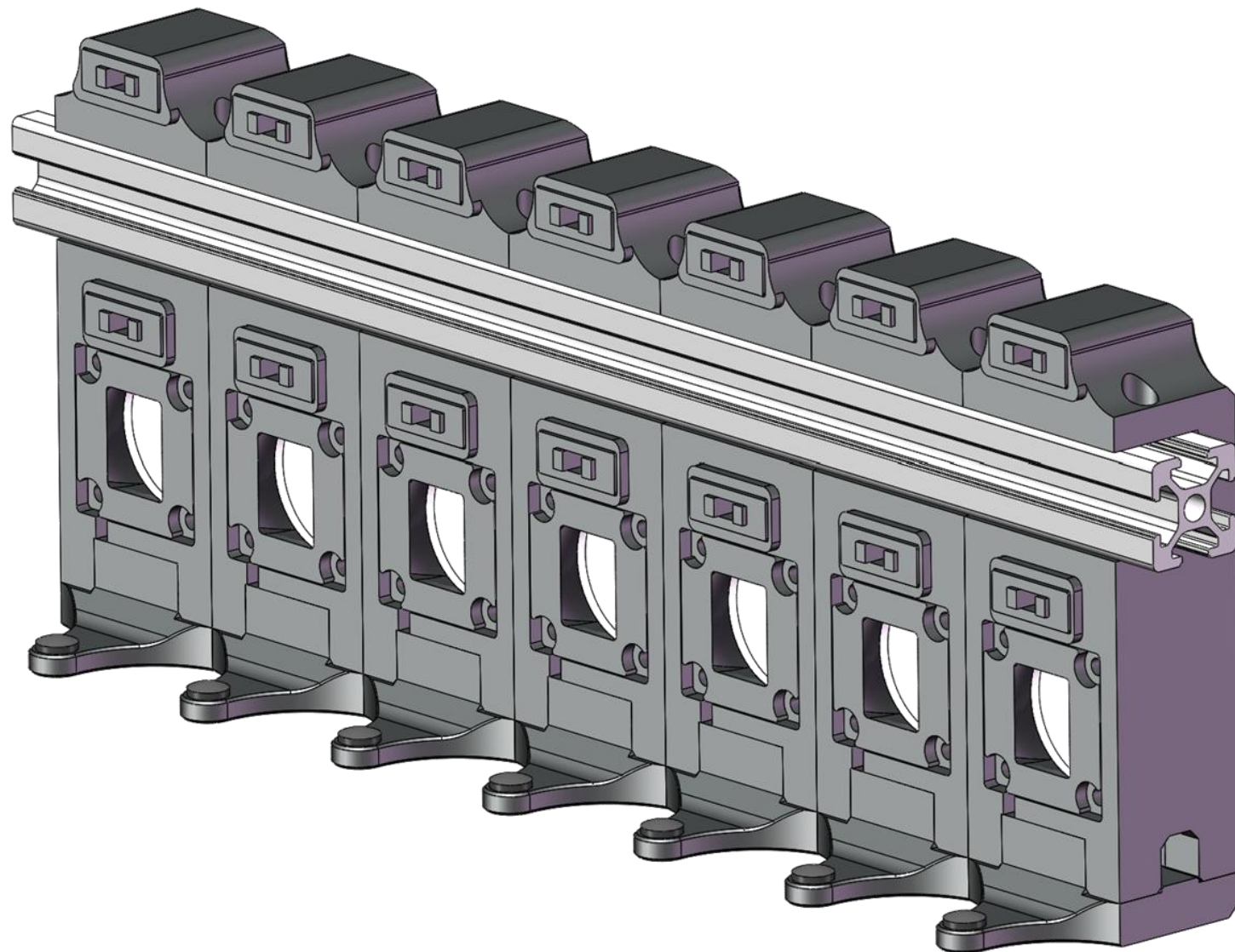
Extend the heater and thermistor wires – the original wires are definitely not long enough.



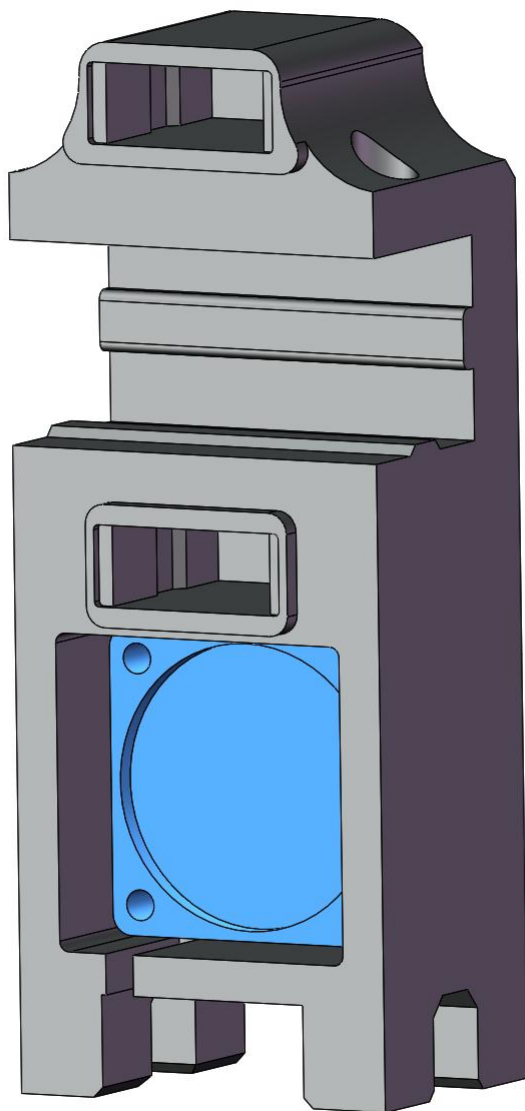
Tool head back plate

As shown, secure with 4 M3 screws. (Screws not drawn for simplicity.)

Part 4: Tool Head Storage Seat Assembly



01 Install the Fan

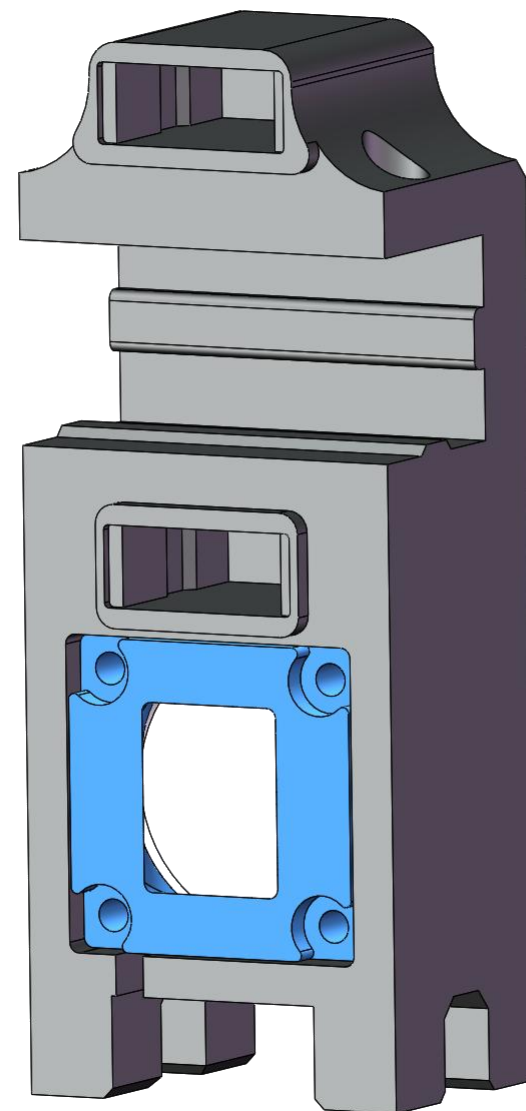


3010fan

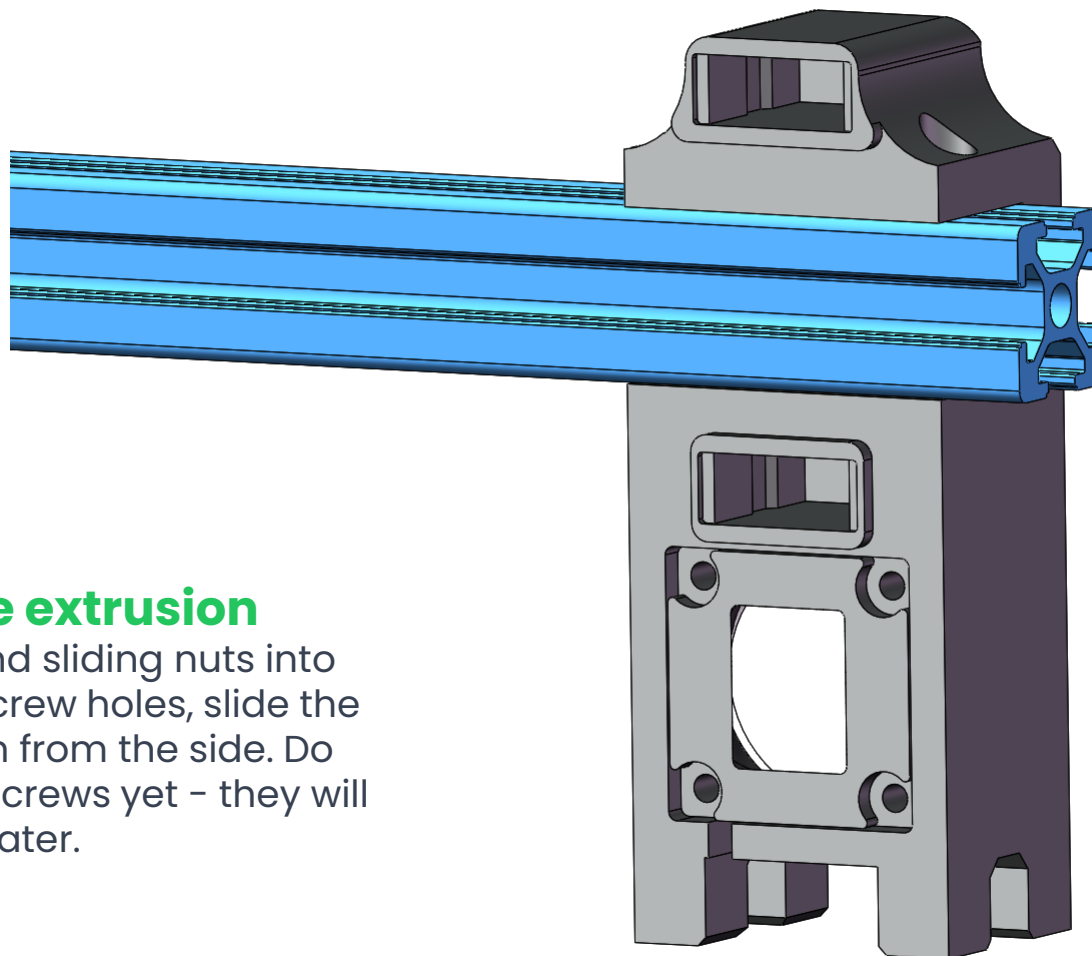
Just clip it in. Pay attention to routing the wire out of the slot.

air guide

Secure with M3 screws.



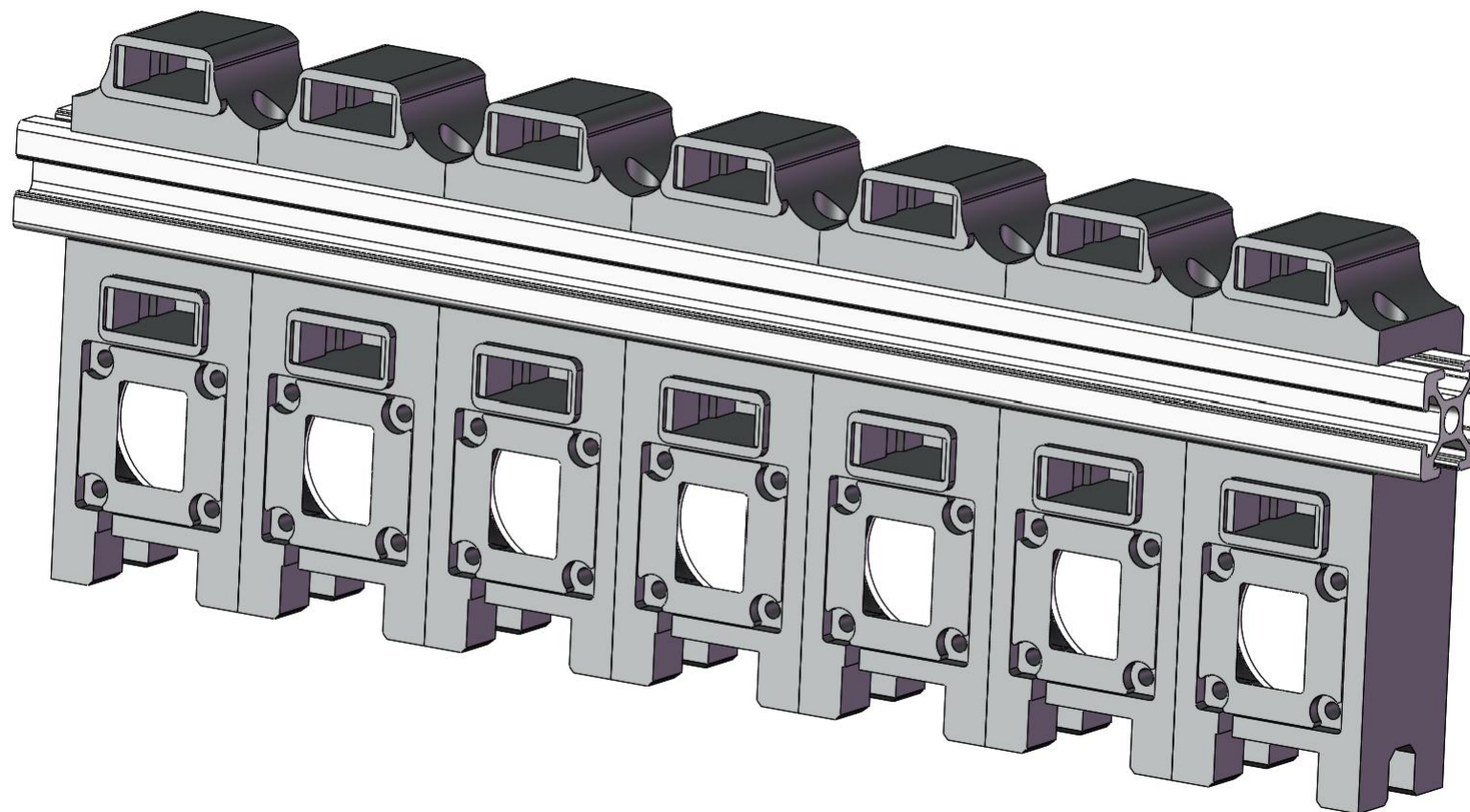
02 Install onto the Extrusion



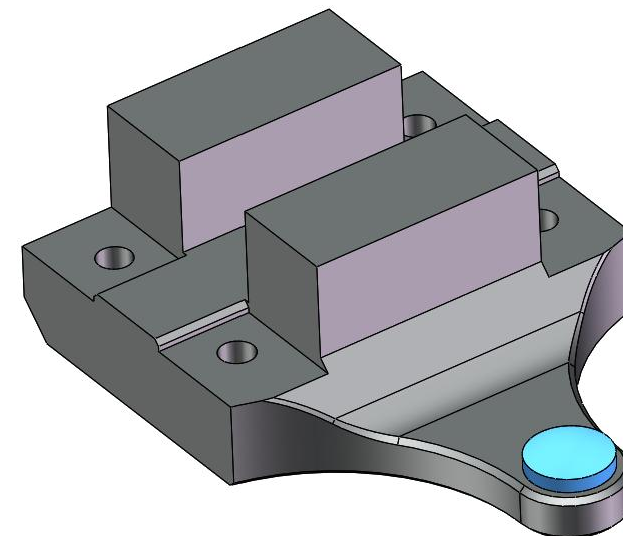
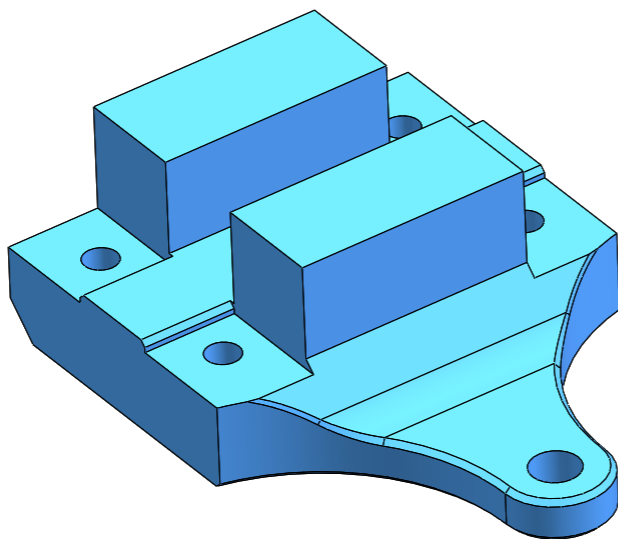
Slide into the extrusion

Screw screws and sliding nuts into the two upper screw holes, slide the tool head seat in from the side. Do not tighten the screws yet - they will need adjusting later.

03



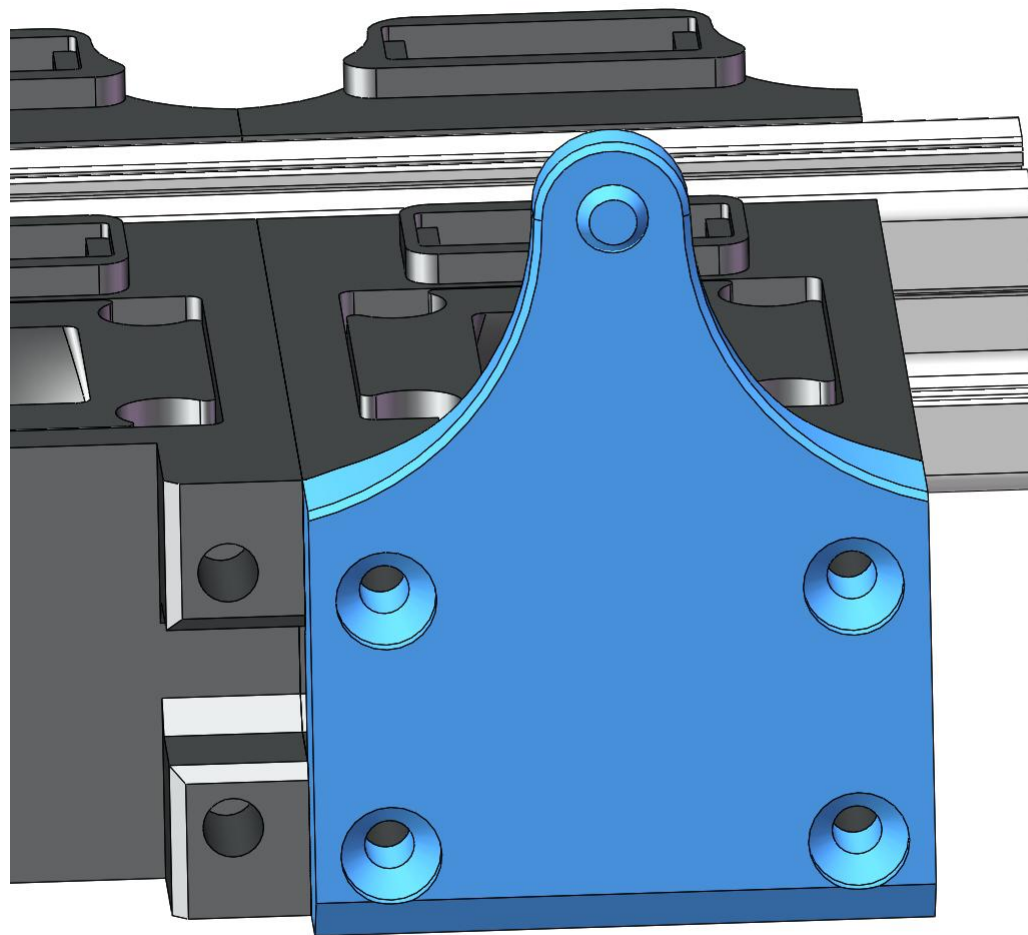
04 Install the Silicone Sleeve



Silicone sleeve

It is quite tight. You can use something pointed, like a hex wrench, to push it in firmly.

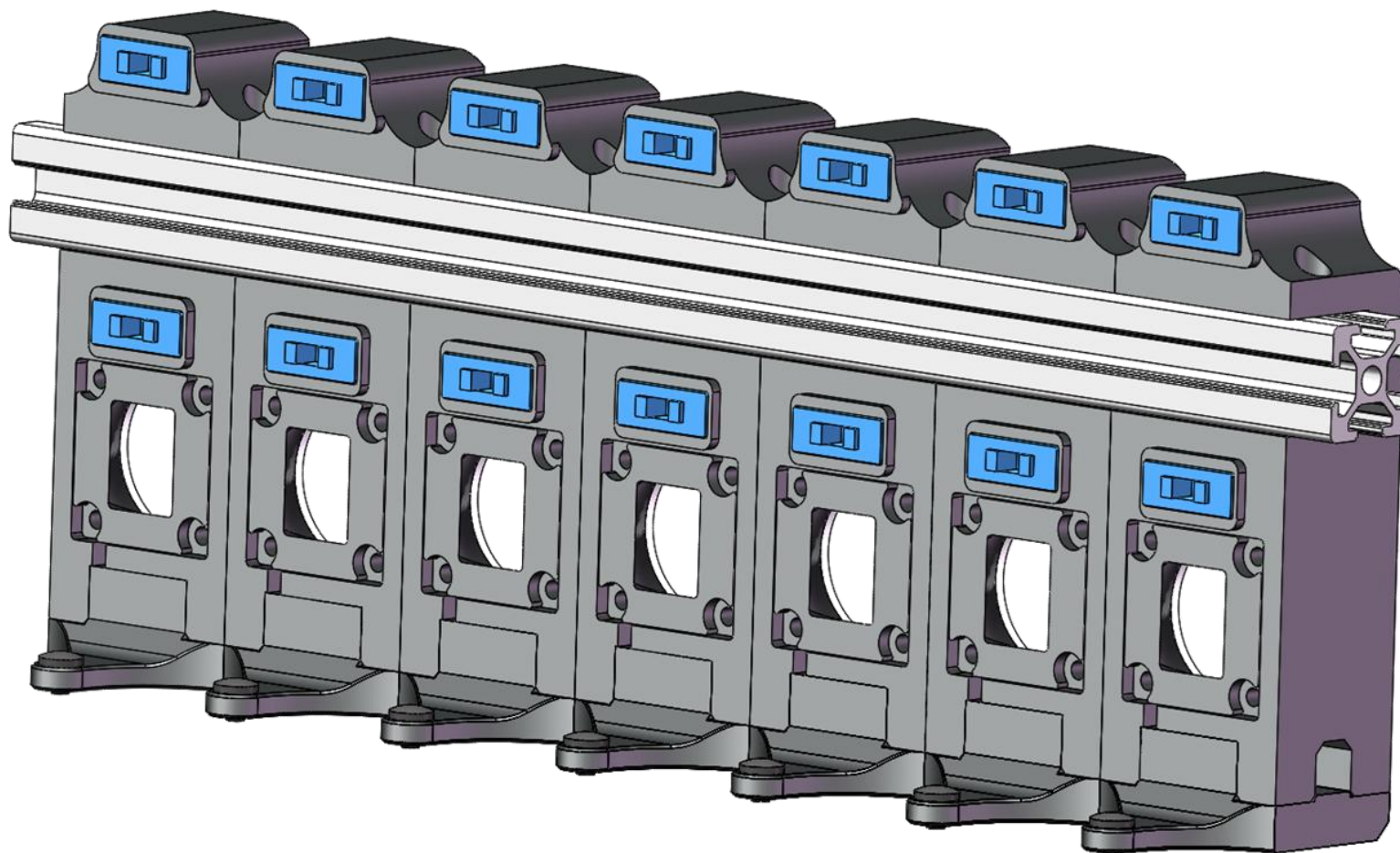
05 Install the Nozzle Cleaning Part



nozzle cleaning part

Tidy up the fan wires, then secure the nozzle cleaning part with 4 screws.

06 Install the Door Lock Switch



Door lock switch

As shown, clip the door lock switch directly in.